

Assurance and Coercion in Conflict

Shusuke Ioku[†]

Shuhei Kurizaki[‡]

September 12, 2026

ABSTRACT. Political actors can fail to secure concessions despite credible threats when targets fear that yielding will invite further coercion. We show that this assurance problem can deepen uncertainty about an opponent's willingness to fight by making demands harder to interpret. When concessions improve the demanding state's military position, states planning an attack can make the same modest demands as states reluctant to fight. We derive this result from a bargaining model in which states choose their demands and retain the option to attack after compliance. Anticipating such attacks, targets can concede less, making settlement less attractive and increasing the risk of war. Yet those planning to attack can gain by securing a smaller concession before launching an attack. Assurance problems can therefore increase conflict while either benefiting or harming the demanding state. Their consequences depend on both what demands reveal and how compliance changes incentives for subsequent violence.

[†]Ph.D. Candidate, Department of Political Science, University of Rochester.

[‡]Associate Professor, Faculty of Political Science and Economics, Waseda University.

INTRODUCTION

Political actors use threats to obtain concessions, but targets may resist credible threats when compliance does not guarantee safety (Schelling 1966). During the Iran nuclear negotiations, Iranian officials feared that an agreement would leave sanctions in place or invite new demands. The Obama administration sought to address these concerns by separating nuclear sanctions from other disputes and managing potential spoilers. Iran accepted the agreement in 2015, but the Trump administration withdrew in 2018 and reimposed sanctions to seek further concessions; Iran then defied the demands and expanded uranium enrichment (Pauly 2024). These events illustrate the general problem of assuring a target that compliance will end the pressure. We study a specific source of that problem: a concession can improve the coercer’s military position before it decides whether to attack.

Our main result concerns what demands reveal about willingness to stop. In equilibria with two positive demands that both receive some compliance, ending the crisis at compliance makes larger demands identify more resolved coercers. Allowing attack after compliance changes this ordering: belligerent and weak coercers can share the low demand while moderate, resolved types demand more and settle after acceptance. Weak types seek a lower risk of resistance; belligerent types seek the military advantage created by compliance. A smaller demand can therefore be less reassuring than a larger one. This ordering is impossible when compliance ends the crisis.

We develop a bargaining model with one-sided private information about the coercer’s cost of war. The coercer chooses a demand, and the target accepts or resists. In the protected benchmark, compliance ends the crisis. Under exposed compliance, the coercer can settle or attack after receiving the concession, and the concession improves its probability of victory. The games otherwise share baseline military power, war costs, and the consequences of resistance.

The same incentive to obtain compliance changes the terms and outcomes of coercion. Comparing the largest pooling demand in each game at a common compliance rate, antici-

pated attack makes the target concede less. The smaller concession reduces settlement value more than the military return to attack, drawing additional types from settlement into war. With two demands, belligerent types can instead obtain compliance by switching to the smaller, more frequently accepted demand. This raises compliance while reducing peaceful transfers in a comparison matching regular feigned restraint to protected monotone sorting at the same demands and acceptance rates, with certain acceptance of the low demand. Exposure raises war risk and conflict costs in both comparisons.

The payoff consequences depend on how bargaining behavior changes. At maximal pooling demands and a common compliance rate, the target's smaller concession leaves the coercer bearing the additional conflict costs. At matched demands and acceptance rates, belligerent types instead gain by obtaining the low-demand concession before attacking. We also allow demands and responses to adjust independently in both games. Under uniform costs, maximal pooling with certain compliance minimizes war and conflict costs and maximizes the coercer's expected payoff and total welfare across the characterized pooling and positive two-demand classes. Exposure worsens all four favorable bounds. These bounds are sharp, although arbitrary equilibrium pairs can reverse the outcome rankings.

First, we contribute to crisis signaling by identifying when a low demand conceals opposite extremes of resolve. In the partially separating equilibrium of [Kydd and McManus \(2017\)](#), aggressive types mimic a moderate type's larger demand and assurance, while a small demand identifies a restrained type. [Slantchev \(2010\)](#) shows that strong types can imitate weak types' smaller demands to preserve tactical surprise in war after rejection; acceptance ends the crisis. Here, compliance supplies the military advantage. Belligerent types choose the low demand to obtain that advantage, weak types choose it to avoid resistance, and moderate resolved types demand more and settle. Protected compliance excludes this ordering within the class of two strictly positive demands receiving compliance. Regular feigned restraint survives every continuous full-support cost distribution. At matched demands and responses, it also makes the demand signal strictly less informative

about binary willingness to fight after resistance, although the signals are not ordered in informativeness about the full war cost.

Second, we extend the analysis of coercive assurance by characterizing how an anticipated attack changes the population willing to settle. [Cebul, Dafoe, and Monteiro \(2021\)](#) emphasize relative power and reputations for restraint, whereas [Pauly \(2024\)](#) emphasizes disentangling demands, managing spoilers, and sharing knowledge. [Kydd and McManus \(2017\)](#) model costly promises of restraint and already allow concessions to shift military power and the target to concede less when it anticipates attack. Their payoff tradeoff also makes smaller concessions reduce settlement value more than the military return to attack. We retain this tradeoff, replace the target’s allocation choice with acceptance or resistance to a demand, and characterize the marginal attacker under continuous private war costs. At maximal pooling demands, the attacking population expands through both the direct military shift and the target’s smaller concession. Retaining the attack option while removing its military benefit makes the two games’ maximal-pooling material outcomes coincide at a common compliance rate. This benchmark separates the effect of concession-induced exposure from changes in which war histories are counted as compliance.

Third, we connect bargaining adjustment to the value of military options. [Schram \(2022\)](#) shows that better limited-attack capabilities can hurt their holder by encouraging greater transgressions, eroding informational gains or inducing its opponent to risk war. [Spaniel and İdrisoğlu \(2024\)](#) show that switching to a cheaper, weaker campaign makes the target less willing to concede and the coercer more willing to risk rejection. Both models end the crisis at accepted agreement; we examine discretion to attack afterward. Holding demands and responses fixed preserves every coercer’s protected opportunities, and belligerent types strictly gain from feigned restraint. Allowing both games to adjust demands and responses establishes a different result: within the characterized classes under uniform costs, exposure lowers the highest attainable coercer payoff and total welfare while raising the lowest attainable war risk and conflict cost. We characterize the exact marginal ranges and construct reversals between individual equilibrium pairs. Thus the loss at the

favorable bound survives bargaining adjustment, while an unrestricted ranking of realized equilibrium outcomes does not.

THE MODEL

Two states bargain over a disputed good valued at 1. The coercer privately observes its cost of war, while the target knows only its distribution. Figure 1 previews their common structure and sole difference: under exposed compliance, the coercer can attack after the target yields. That difference will determine whether a low demand can conceal a belligerent coercer.

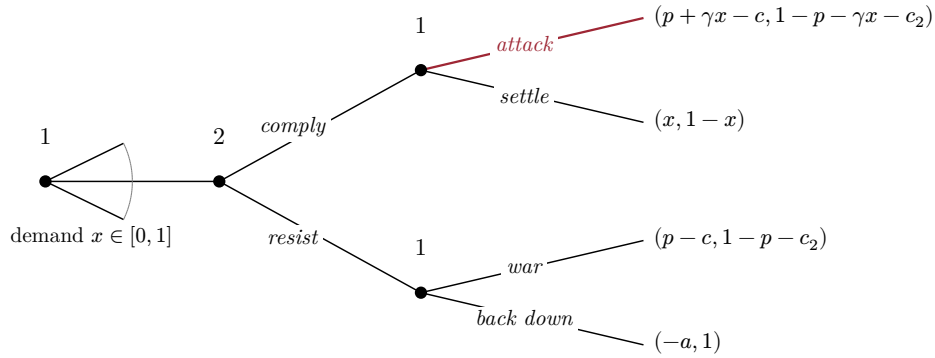


Figure 1: Protected and exposed compliance in one game tree

Note: Player 1 is the coercer and player 2 the target; the coercer's payoff appears first. The tree is conditional on the privately observed cost c . Nature's draw is omitted; target nodes following the same demand across different cost realizations belong to one information set. Under protected compliance, yielding ends at settlement. Under exposed compliance, the red branch permits attack after yielding, and a concession x raises the coercer's probability of victory from p to $p + \gamma x$.

Actions and payoffs

The coercer's war cost c is uniformly distributed on $[0, \bar{c}]$, with cumulative distribution function F . The target's war cost $c_2 \geq 0$ is common knowledge. Ordinary war gives payoffs

$$(p - c, 1 - p - c_2)$$

where $p \in (0, 1)$ is the coercer's probability of victory. Let $h = p + c_2$ denote the target's loss from war.

The coercer demands $x \in [0, 1]$. The target observes the demand and either complies, transferring x , or resists. After resistance, the coercer chooses war or backing down. Backing down imposes an audience cost $a > 0$ and gives $(-a, 1)$. The coercer fights after resistance if and only if

$$c < r \equiv p + a \quad (1)$$

Under protected compliance, yielding ends play at $(x, 1 - x)$. We mark outcomes in this benchmark with the subscript P . Under exposed compliance, the coercer chooses after yielding between settlement at $(x, 1 - x)$ and attack. We mark outcomes in this comparison with the subscript E . Compliance shifts a fraction γ of the concession into the coercer's probability of victory, so attack gives

$$(p + \gamma x - c, 1 - p - \gamma x - c_2) \quad (2)$$

The coercer attacks after compliance if and only if

$$c < t_E(x) \equiv p - (1 - \gamma)x \quad (3)$$

The parameter γ is the share of each conceded unit converted into military advantage. The shift is zero when the target transfers nothing and increases with the concession. Because $\gamma < 1$, a larger transfer still makes settlement more attractive relative to attack. War reopens the whole dispute: the transfer is not retained separately after attack, but its effect on the balance of power persists in the war lottery. This specification captures concessions such as demobilization, surrendered terrain, or access that alter the target's military position.

We solve for perfect Bayesian equilibria (PBEs) satisfying the Intuitive Criterion ([Cho and Kreps 1987](#)). After an unused demand, the criterion excludes types that cannot attain their equilibrium payoff under any sequentially rational target response, provided some type remains. [Lemma 3](#) supplies supporting beliefs and responses for every equilibrium

construction without changing on-path behavior. All equilibrium claims below impose this requirement.

In pooling equilibria, a range of types shares one accepted demand. In separating equilibria, used here as shorthand for two-demand sorting, different groups choose different accepted demands. Equilibrium classes count only demands used on path. The complete sorting characterization covers two strictly positive demands, each chosen by a positive mass of types and receiving positive compliance. Zero-demand boundaries are described separately.

Let $\lambda(x)$ denote the target's probability of compliance after demand x , and write it as λ_j for the accepted pool in game $j \in \{P, E\}$. This probability is chosen by the target. Both maximal pooling demands permit any $\lambda_j \in (0, 1]$ through target indifference; the pooling outcome comparison matches $\lambda_P = \lambda_E = \lambda$. The Intuitive Criterion preserves this range of responses without selecting one. At continuation ties the coercer settles after compliance and backs down after resistance. Such ties have probability zero in the finite-demand profiles studied here. Reassurance means settlement with probability one conditional on compliance.

ASSUMPTION 1 (Interior benchmark): The parameters satisfy $h < 1$, $r < \bar{c}$, $\bar{c} > 2p + c_2$, and $hr < p\bar{c}$.

ASSUMPTION 2 (Concession-dependent exposure): The exposure rate satisfies $0 < \gamma < 1$ and $p + \gamma < 1$.

Assumption 1 gives an interior protected-compliance equilibrium. Assumption 2 keeps the probability of victory below one for every feasible demand. It also ensures that the concession's settlement value rises faster than the military exposure it creates.

Kydd and McManus (2017) let the target choose a continuous allocation after observing the coercer's goal and any threat or assurance. Here, the target accepts or resists a demand,

and the coercer’s war cost is continuously distributed and privately known. We retain their allocation-dependent military payoff and the tradeoff between settlement value and the incentive to attack. The continuum lets us characterize both the marginal attacking type and the intervals of types choosing each demand. The two-demand analysis identifies the change in incentives: the relative value of the high demand falls monotonically with war cost under protection, but can rise and then fall when compliance permits attack.

Compliance changes military power before the coercer chooses whether to attack, and the target cannot reverse that change within the game. Attack after compliance carries no separate reputational penalty. The comparison therefore couples the availability of post-compliance attack with strictly positive military exposure. Costly promises and reputations for restraint are other ways to alter the incentive to stop (Kydd and McManus 2017; Cebul, Dafoe, and Monteiro 2021); here, reassurance is a property of the coercer’s continuation behavior rather than a separate message.

Timing

Nature draws c and reveals it to the coercer. The coercer makes a demand, the target complies or resists, and play reaches the continuation outcome shown in Figure 1.

Equilibrium Outcomes and Measures

Equilibria produce three substantively different outcomes. Coercion can fail because every demand is resisted. If some demand receives compliance, that response either ends the crisis or is followed by attack with positive probability. Among equilibria built around two demands that both receive some compliance, the games differ sharply: protected compliance restricts the low demand to less resolved types, whereas exposed compliance also lets belligerent types hide behind it.

These possibilities explain the structure of the analysis. When every demand is resisted, coercion fails and assurance has no role because the target never entrusts its security to the coercer’s restraint. Once some demand is accepted, assurance becomes consequential:

the same act of compliance can either secure peace or leave the target exposed to attack. That difference determines how much the coercer can obtain peacefully, how often conflict occurs, and how its costs are distributed. We therefore organize the results around coercive ability, war risk, and welfare.

DEFINITION 1 (Coercive outcomes): Let C denote compliance, S settlement after compliance, W either ordinary war or post-compliance attack, and D the on-path demand. For $\Pr(C) > 0$, *coercive ability* is $\Omega = \frac{E[D\mathbb{1}[S]]}{\Pr(C)}$: the peaceful concession obtained per instance of compliance. The *expected peaceful transfer* is $T = E[D\mathbb{1}[S]]$, total war risk is $q = \Pr(W)$, expected conflict cost is $K = E[(c + c_2)\mathbb{1}[W]]$, and the coercer's expected payoff is U . Write $I(z)$ for the expected conflict cost generated by types below $z \in [0, \bar{c}]$. If the target's expected payoff is U_2 , total welfare is $V = U + U_2$.

The distinction between Ω and T matters when the target mixes. A concession can be valuable conditional on receiving it but rare ex ante. When coercion fails, $T = 0$ and Ω is undefined.

A zero-exposure benchmark further separates compliance histories from material outcomes. Retain the attack option but set $\gamma = 0$. At maximal pooling demands and a common compliance rate, the games then have identical peaceful transfers, war probabilities, conflict costs, and payoffs. Low-cost types may nevertheless accept and attack under exposure while fighting after resistance under protection. Counting their acceptance raises compliance and lowers Ω without changing any material outcome. At this boundary, their decision to join the accepted pool is indifferent, so the composition effect varies across exposed pools. [Lemma 6](#) establishes the equivalence and its scope.

EQUILIBRIUM CONSEQUENCES OF ASSURANCE

Equilibria in which every demand is rejected add no assurance effect because compliance never occurs. The analysis focuses on positive accepted demands. With one accepted demand, the target cannot distinguish among the types sharing it. With two strictly positive accepted demands, types can sort either monotonically by resolve or with belligerent and weak types using the low demand around a moderate high-demand group. Apart from types that are indifferent between the demands, these two orderings exhaust the class with pure demand strategies, two strictly positive demands, and positive compliance at both. Call separation *reassuring* when compliance with either accepted demand ends in settlement almost surely. Complete strategies, beliefs, and proofs appear in the Supporting Information.

PROPOSITION 1 (Loss of assurance narrows reassuring separation and permits feigned restraint): Under Assumptions 1–2:

1. **Pooling.** Both games admit positive-compliance pooling throughout the parameter space.
2. **Separating equilibria.** With two strictly positive demands, protected compliance admits reassuring monotone sorting and excludes feigned restraint. Exposed compliance admits feigned restraint throughout, and reassuring monotone sorting if and only if $\gamma \leq \frac{c_2}{h}$.

The proposition separates the availability of coercion from the information conveyed by a demand. Removing assurance does not eliminate pooling. It instead narrows the conditions under which two demands can separate types while assuring peace and creates a new form of separation in which belligerent types can feign restraint with a low demand.

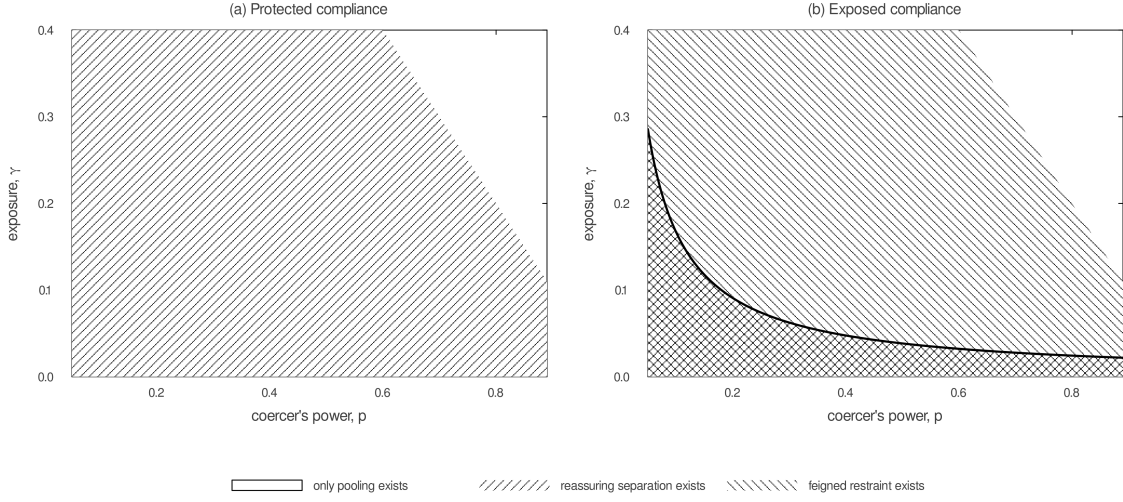


Figure 2: Exposure narrows reassuring separation and permits feigned restraint

Note: Pooling equilibria exist at every admissible point in both panels. Within the admissible region, white fill denotes parameters at which only pooling exists, while hatching marks separating equilibria with two strictly positive demands. Under protected compliance, reassuring separation exists throughout, and feigned restraint is impossible. Under exposed compliance, feigned restraint exists throughout for $\gamma > 0$, while reassuring separation survives on and below the solid boundary $\gamma = \frac{c_2}{h}$; crosshatching indicates that both forms of separation exist. The white upper-right wedge lies outside the maintained parameter space because it violates $p + \gamma < 1$. Fixed parameters are $c_2 = 0.02$, $a = 0.30$, and $\bar{c} = 2.00$.

Coercive Ability

Protected compliance lets resolved coercers make a high demand without making the target fear an attack after acceptance. Let x_P be the largest accepted demand within the one-accepted-demand pooling class. Types with sufficiently low war costs prefer to make a demand that will be resisted and then fight. The remaining types demand x_P and settle when the target complies. Because every accepted demand ends peacefully, x_P is also the amount peacefully coerced conditional on compliance. In the pooling comparison, each game's accepted demand receives the same compliance probability $\lambda \in (0, 1]$.

Exposure changes both the composition of the accepted pool and its maximal demand. If a positive demand receives any chance of compliance, a belligerent type gains from joining that pool: it can take the concession, improve its military position, and attack.

Belligerent types therefore cannot remain in a separate pool whose demand is always resisted. Anticipating their entry, the target accepts a smaller maximal pooling demand, $x_E < x_P$.

The smaller demand also expands the set of types that fight regardless of the target's response. Let t_P mark the protected types that choose the refused demand and t_E the exposed types that attack after compliance. These are the war cutoffs when the target complies surely with the accepted demand. The two changes are linked:

$$t_E - t_P = \underbrace{\gamma x_E}_{\text{direct military shift}} + \underbrace{(x_P - x_E)}_{\text{demand feedback}} > 0 \quad (4)$$

The first term is the military advantage created directly by the concession. The second is the bargaining response: fear of attack makes the target concede less, which makes settlement less attractive to additional coercer types. Because both terms are positive, the target's defensive response amplifies rather than offsets the loss of assurance.

At the maximal pooling demand in each game, the target is exactly indifferent between complying and resisting. Its compliance rate can therefore vary across equilibria. Matching that rate across games, exposure lowers both conditional coercive ability and the expected peaceful transfer for every $\lambda > 0$. Greater exposure pushes the accepted demand down further and moves more types from settlement into attack. Military power still improves coercive ability in both games over the relevant range, but it improves it less when compliance remains exposed: some of the extra leverage is lost to the target's smaller concession and to attack after compliance.

Separation through two strictly positive demands restricts peaceful coercion in two further ways. Focus on equilibria with $0 < x < y$ in which the target sometimes complies with each demand. The low demand must be accepted more often. In *monotone sorting*, lower-cost types choose the high demand and higher-cost types choose the low demand. In the second pattern, belligerent types use the low demand to feign restraint: intermediate types choose the high demand while both tails choose the low demand. The Supporting

Information calls this pattern *two-sided sorting*. Apart from types exactly indifferent between the demands, these patterns exhaust this class.

In monotone sorting, lower-cost coercers risk resistance to obtain the larger concession, while higher-cost coercers choose the low demand because it is accepted more often. Protected compliance sustains $y = h$ because all high-demand types fight after resistance and settle after compliance. Exposure preserves that demand only when every high-demand type still settles. Once attackers enter the pool, the target accepts the high demand only if it falls below h . A high demand then reveals willingness to fight without assuring the target that yielding will end the crisis.

Regular feigned restraint is unique to exposed compliance. In this class, exposure gives the lowest-cost types another reason to value compliance: they choose the low demand, obtain its higher probability of acceptance, and then attack from an improved position. Weak types choose the same demand to avoid resistance. Moderate resolved types instead make the high demand, fight if it is resisted, and settle if it is accepted. Under protected compliance, the return to acceptance changes monotonically with resolve, so the two tails cannot pool around an interior interval. Exposure changes that ordering: the low demand can conceal belligerence, while the high demand identifies resolved types willing to settle.

Regular feigned restraint survives replacing uniform costs with any continuous, strictly increasing distribution on the same support. The cutoffs retain their pointwise incentive conditions; the target's acceptance constraint instead weights the demand groups by their probabilities under the new prior. The Supporting Information proves existence for every positive exposure rate and shows that the matched outcome signs below survive. A particular menu need not remain feasible when the prior changes. The pooling comparative statics and attainable bounds retain the uniformity assumption.

Feigned restraint also occurs with substantial demands and acceptance rates. In the interior example in [Supporting Information O](#), the low demand is 0.20 and is accepted for certain; the high demand is 0.62 and is accepted with probability 0.20. Belligerent types

comprise about 19 percent of the population, and the high-demand group comprises 15 percent. These demands, acceptance rates, and group masses remain bounded away from zero in a neighborhood of the example.

The model's distinguishing implication joins demand size to the outcome after acceptance. In regular feigned restraint, compliance with the low demand sometimes precedes attack, whereas compliance with the high demand ends in settlement. Measuring compliance alone therefore misses the difference between obtaining an agreement and extracting a peaceful concession.

In the exposed equilibrium used for comparison, types below b are the belligerent members of the low-demand pool, types between b and k make the high demand and settle if it is accepted, and types above k make the low demand to avoid resistance. Let A denote attack after compliance. The probability that a complied-with low demand is followed by attack is

$$\Pr(A \mid x, C) = \frac{b}{b + \bar{c} - k} \quad (5)$$

where the numerator and denominator are the lengths of the attacking interval and the entire low-demand pool. Greater exposure raises b while leaving k fixed when the demands and their relative acceptance rates are held fixed. The low demand thus contains more attackers, becomes less reassuring, and is harder to sustain. The target accepts it surely while the gains from peaceful settlement with resolved types in the low-demand pool strictly exceed the losses from conceding to weak and belligerent types. If that compensation becomes insufficient, the specified demand and response profile ceases to be an equilibrium. Other feigned-restraint equilibria remain available.

For the outcome comparison, take a regular exposed feigned-restraint equilibrium in which the low demand is accepted for certain and the high demand is accepted with probability ρ . Match it to a protected monotone equilibrium with the same demands and target responses. Under protection, every resolved type below k uses the high demand and

settles if it is accepted. Under exposure, types below b move to the low demand and attack after compliance. Let P_C denote the ex ante probability of compliance. Exposure changes compliance and the peaceful transfer by

$$\Delta P_C = \frac{(1-\rho)b}{\bar{c}} > 0, \quad \Delta T = -\frac{\rho hb}{\bar{c}} < 0 \quad (6)$$

The first comparison says that compliance becomes more common because the belligerent types switch to the low demand, which is accepted more often. The second says that peaceful concessions nevertheless decline because those same types no longer settle after their high demand is accepted. The high-demand settlement set contracts from $[0, k)$ to $[b, k)$. When exposure is too limited to induce post-compliance attack, the exposed monotone equilibrium reproduces the protected outcome.

This change in demand choice has a precise information consequence. At the fixed prior, demand under regular feigned restraint is strictly less informative about whether the coercer would fight after resistance. The protected signal can reproduce the exposed signal by randomly relabeling some high demands as low, leaving low demands unchanged. Exposure thereby makes some resolved coercers indistinguishable from weak coercers. The comparison concerns binary resistance resolve: the experiments are incomparable about the full war cost, because the exposed high demand distinguishes an intermediate group that the protected signal pools with the most belligerent types. [Lemma 5](#) gives the exact comparison.

PROPOSITION 2 (Loss of assurance reduces coercive ability):

1. **Pooling.** At the maximal pooling demands and every common compliance rate $\lambda > 0$, $\Omega_E < \Omega_P$ and $T_E(\lambda) < T_P(\lambda)$.
2. **Separating equilibria.** Matching regular exposed feigned restraint to protected monotone sorting at the same demands and target responses, with certain low-demand acceptance, $\Omega_E < \Omega_P$ and $T_E < T_P$.

For pooling, power raises coercive ability in both games when $p < \bar{p} = \frac{\bar{c}-c_2-a}{2}$, but raises it less under exposed compliance. Exposure thus reduces both the level of peaceful coercion and its responsiveness to power over this range.

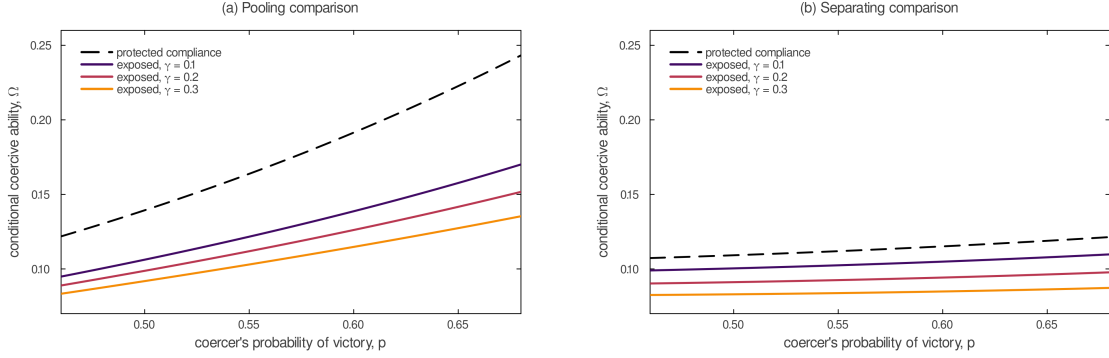


Figure 3: Loss of assurance reduces coercive ability in pooling and separating equilibria

Note: Both panels vary the coercer's probability of victory and plot conditional coercive ability Ω . The black dashed line is protected compliance; the solid lines use a dark-to-light gradient for exposed compliance at $\gamma = 0.10, 0.20$, and 0.30 . Panel (a) uses the maximal pooling demand in each game and pure compliance with that demand in both games. Its exposed curve is $x_E[1 - F(t_E)]$, rather than the demand alone. Panel (b) compares protected monotone equilibria with exposed equilibria in which the belligerent types feign restraint with a low demand. Parameters are $c_2 = 0.02$, $a = 0.30$, and $\bar{c} = 2$ in both panels, with $x = 0.10$ and $\rho = 0.08$ in panel (b).

War Risk

At maximal pooling demands and a common compliance rate, exposure raises war risk. Types below t_P fight in both games. Types between t_P and t_E settle after compliance under protection but attack after compliance under exposure. They already fight after resistance in either game. Exposure therefore adds war with probability λ for exactly those types. Types between t_E and r continue to settle after compliance and fight after resistance in both games.

In the matched separating comparison, only the belligerent types below b change signals. They settle when their high demand is accepted under protection. Under exposure, they obtain the low-demand concession and then attack. Their switch therefore adds war without preventing any war elsewhere in the type distribution.

PROPOSITION 3 (Loss of assurance raises war risk and conflict costs):

1. **Pooling.** At the maximal pooling demands and every common compliance rate $\lambda > 0$, $q_E(\lambda) > q_P(\lambda)$ and $K_E(\lambda) > K_P(\lambda)$.
2. **Separating equilibria.** In the matched comparison, exposure raises war risk by $\frac{\rho b}{c}$ and expected conflict cost by $\rho I(b)$.

The increase $\rho \frac{b}{c}$ has a direct interpretation: $\frac{b}{c}$ is the population share that loses the peaceful high-demand option, and ρ is the probability with which that option previously ended in settlement. The associated cost increase, $\rho I(b)$, sums the war costs of exactly those types. In pooling equilibria, war risk and conflict cost vary with the target's response in both games. The figure reports pure compliance with each accepted demand, which minimizes these quantities within each maximal-demand family.

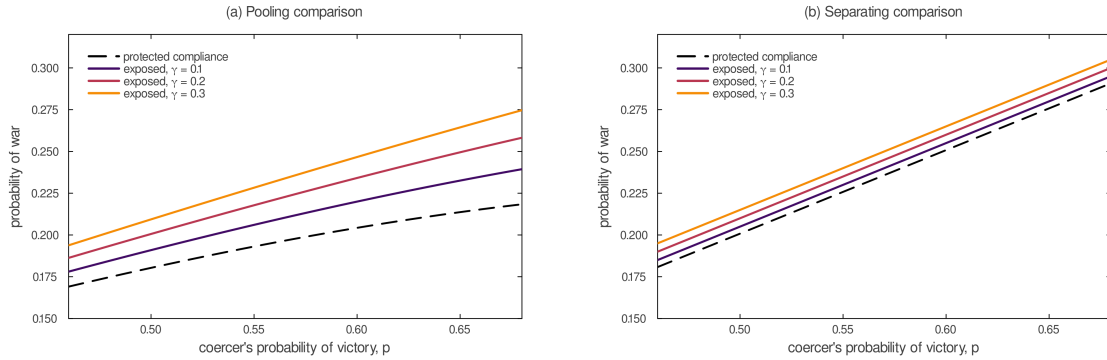


Figure 4: Loss of assurance raises war risk in pooling and separating equilibria

Note: Both panels vary the coercer's probability of victory and use the same axes, exposure rates, and line meanings as Figure 3. Panel (a) shows the maximal-demand pooling comparison under pure compliance with the accepted demand in both games, which minimizes war risk within each family. Panel (b) compares protected monotone sorting with the feigned restraint available only under exposed compliance. In the exposed game, belligerent types obtain compliance with a low demand and then attack.

Welfare

War imposes costs on both sides, so the additional wars in the matched comparisons lower total welfare. At maximal pooling demands, resistance also imposes an audience cost when

the coercer backs down. A common compliance rate gives the same expected audience cost in both games, so the welfare difference equals the additional conflict cost. The target is held to its payoff from resistance at either maximal pooling demand. Its expected payoff is therefore unchanged, leaving the coercer to bear the entire welfare loss.

The matched feigned-restraint comparison holds the demands and the target's response to each demand fixed. Every coercer can obtain at least its protected continuation payoff, because the attack option adds a choice without removing settlement. The substantive result is that the same demands and responses can be sustained in equilibrium while belligerent types below b strictly gain by switching to the low demand and attacking after compliance. The target loses the coercer's net payoff gain plus the added joint conflict costs.

PROPOSITION 4 (Loss of assurance lowers total welfare):

1. **Pooling.** At the maximal pooling demands and every common compliance rate $\lambda > 0$, $V_E(\lambda) < V_P(\lambda)$. The target's expected payoff is unchanged, so the coercer bears the welfare loss: $U_E(\lambda) < U_P(\lambda)$.
2. **Separating equilibria.** In the matched comparison, total welfare falls by $\rho I(b)$. The coercer gains $\frac{\rho b^2}{2\bar{c}}$, but the target loses $\frac{\rho b(b+c_2)}{\bar{c}}$.

The coercer's gain, $\rho \frac{b^2}{2\bar{c}}$, is the net benefit to the belligerent types that change demands. The target loses that amount plus their added conflict costs, so total welfare falls by $\rho I(b)$. In pooling, the target instead reduces the accepted concession and preserves its payoff from resistance. The two comparisons therefore distinguish direct benefits at given demands and responses from losses caused by the target's bargaining adjustment.

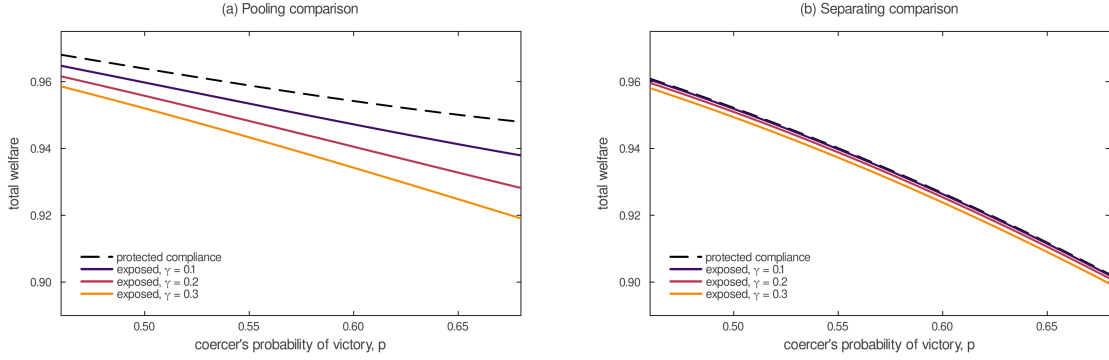


Figure 5: Loss of assurance lowers total welfare in pooling and separating equilibria

Note: Both panels vary the coercer's probability of victory and plot total welfare V , using the exposure rates and line meanings in Figure 3. Panel (a) shows the maximal-demand pooling comparison under pure compliance with the accepted demand in both games, which maximizes welfare within each family. Panel (b) compares protected monotone sorting with the feigned restraint available only under exposed compliance. The exposed game adds conflict among the belligerent types that use the low demand.

Bargaining Adjustment

The preceding comparisons either match compliance rates at maximal pooling demands or hold a two-demand menu and its responses fixed. We now let both games adjust demands, group composition, and target responses. The question is which outcomes can be attained across the characterized classes, including exposed monotone sorting and profiles sustained by demand indifference.

PROPOSITION 5 (Exposure worsens the favorable bounds of attainable outcomes):

Under uniform costs, maximal pooling with certain compliance minimizes war risk and conflict costs and maximizes the coercer's expected payoff and total welfare across the characterized pooling and positive two-demand classes. The bounds are strict for every positive two-demand equilibrium and sharp within that class.

Exposure strictly raises both minima and lowers both maxima.

The result uses the same comparison rule in both games. Their favorable bounds are $q_j^* = F(t_j)$, $K_j^* = I(t_j)$, $U_j^* = hF(r) - I(t_j)$, and $V_j^* = 1 - I(t_j)$. Since $t_E > t_P$, exposure

worsens all four. The target's payoff at either favorable endpoint equals its resistance payoff. The loss in the maximum coercer payoff thus survives allowing all characterized two-demand strategies and target responses to adjust. This is an *ex ante* bound; it does not rank the payoffs of every coercer type.

Two-demand bargaining cannot improve these bounds because the target must be willing to accept each demand. With certain low-demand acceptance, positive acceptance of the high demand spreads war over a larger range of costs instead of concentrating it among the lowest-cost types. Under the uniform prior, the coercer's payoff envelope links this additional conflict cost to the low demand it can sustain. Target participation then keeps that demand below the maximal pooling demand. Reducing low-demand acceptance mixes these outcomes with universal resistance and cannot improve the favorable bounds.

The Supporting Information gives the exact marginal ranges. Each exposed range is strictly contained in its protected counterpart, with universal resistance providing their common opposite endpoint. The ranges describe one outcome at a time and do not imply that any combination of their values is attainable. Individual equilibrium pairs can still reverse every one of the four rankings, even when both low demands are accepted certainly. The Intuitive Criterion supports the characterized profiles but selects neither their favorable bounds nor a particular pair. The result covers pooling and pure demand strategies with two strictly positive demands, positive mass at both, and positive compliance at both; zero-demand sorting and arbitrary signaling strategies remain outside its scope.

CONCESSION AND EXPOSURE AT MUNICH

Munich illustrates the distinction between a limited claim and a willingness to stop. Hitler presented the Sudeten claim as his final territorial demand in Europe on 26 September 1938, yet his military directive of 30 May had already contemplated the destruction of Czechoslovakia and occupation of Bohemia and Moravia. The contrast makes it difficult to attribute Germany's broader aims solely to developments after the settlement. A limited settlement nevertheless remained a publicly plausible interpretation: on 28 September,

Chamberlain described Hitler’s assurances about implementation as credible. These records support an interpretation of concealed aims and uncertainty about intentions, features of the crisis already emphasized by [Kydd and McManus \(2017\)](#). They do not show that the target trusted Germany to stop.^[1]

The concession also transferred means of defense. The Munich Agreement required evacuation to be completed without destroying existing installations, and a contemporary US report relayed the Czech legation’s account of fortifications and armaments passing to Germany. The military consequences had been anticipated: a US attaché had warned that the Godesberg map would leave Czechoslovakia effectively unable to defend itself. On 21 October, a German directive envisaged occupation of the remaining Czech territory with substantially smaller preparations than Plan Green, although it made action conditional on Czech policy becoming hostile. Together, these records support the concession-induced exposure central to the model; they do not measure how much of the subsequent military advantage came from the territorial transfer.^[2]

Acceptance did not require confidence in German restraint. Before Munich, the US ambassador in Paris reported that Bonnet expected Czechoslovakia to yield when Britain and France made clear that it could expect no assistance. [Fang, Johnson, and Leeds \(2014\)](#) explain Czech compliance through the importance of French support and the costs of defying an ally’s recommendation. British officials could also favor delay while recognizing that German expansion would strengthen Germany’s military position. These were different incentives borne by different states: the four powers negotiated the agreement, while Czechoslovakia surrendered the territory. The case therefore cautions against treating a

^[1]Hitler’s directive of 30 May 1938, PS-388, item 11, §§1–5, [Harvard Nuremberg Trials Project](#); Hitler’s speech of 26 September 1938, [French Yellow Book](#), no. 11; Chamberlain’s statement, [House of Commons](#), 28 September 1938.

^[2]Munich Agreement, 29 September 1938, arts. 1–4; Carr to the Secretary of State, 27 September 1938, [FRUS 1938](#), I, doc. 655; Wilson to the Secretary of State, 6 October 1938, doc. 698; directive of 21 October 1938, C-136, §2, [Harvard Nuremberg Trials Project](#).

broker's reassuring language as the target's reason for compliance.^[3]

The demand-choice mechanism requires an additional link that Munich does not establish. At Godesberg, Hitler rejected the proposed procedure for an already accepted territorial transfer and imposed more demanding terms; Kydd and McManus also report that he preferred war to a negotiated settlement. Neither observation establishes a decision to lower the demand in order to obtain compliance. Moreover, the March 1939 occupation followed a fresh ultimatum and an order not to resist, limiting its correspondence to the model's costly war after acceptance. Munich documents concealed expansive aims and the loss of defenses through compliance, but it does not establish that belligerent and weak coercers choose the same low demand while moderate, resolved coercers demand more and settle.^[4]

The Supporting Information records the evidence and its measurement limits. The distinguishing observable implication remains a joint one: the same low demand must attract both actors seeking to avoid resistance and actors seeking the military benefit of compliance, while a higher demand attracts actors willing to stop. A concession followed by further coercion establishes neither that sorting nor the reason the lower demand was chosen.

EQUILIBRIUM SCOPE

Both games also permit complete coercive failure: pessimistic beliefs can support equilibria in which every demand is resisted. Assurance has no on-path effect in these cases because the target never complies. The Supporting Information provides a construction.

^[3]Bullitt to the Secretary of State, 25 September 1938, [FRUS 1938, I, doc. 623](#); Ismay's note, 20 September 1938, CAB 21/544, para. 15, reproduced by [The National Archives](#), pp. 11–12; Carr to the Secretary of State, 7 November 1938, [FRUS 1938, I, doc. 710](#).

^[4]Chamberlain's statement, House of Commons, 28 September 1938; (Kydd and McManus 2017), pp. 343–344; German record of the Hitler-Hácha meeting, 15 March 1939, PS-2798, pp. 437–439, [Harvard Nuremberg Trials Project](#); Carr's account transmitted by Linnell, 25 March 1939, [FRUS 1939, I, doc. 58](#).

The imitation result applies beyond the equilibria used for outcome comparisons. Whenever a belligerent type could imitate a positive demand receiving positive compliance and then attack, it cannot remain in an always-resisted pool. The Supporting Information also proves that sorting between two strictly positive demands has only the two orderings described above, apart from indifferent types. Protected compliance rules out feigned restraint within this class. Zero-demand boundaries require separate treatment, and the analysis does not characterize arbitrary signaling strategies.

The matched comparisons identify how exposure changes behavior under specified bargaining adjustments. The attainable bounds answer a separate question using a common rule across games: even the best outcomes within the characterized classes deteriorate under exposure. Neither result provides an unrestricted ordering of individual equilibria. The Supporting Information constructs reversals with strictly positive demands and certain low-demand compliance, and shows that changing compliance rates can reverse pooling comparisons as well. At zero exposure, maximal-pooling material outcomes agree even though compliance-conditioned measures can differ. These qualifications identify which conclusions concern military exposure, which concern bargaining adjustment, and which depend on how compliance is counted.

CONCLUSION

A demand reveals willingness to stop only in light of what compliance allows the coercer to do. With two positive demands that both receive some compliance, exposed compliance permits belligerent and weak types to share the low demand while moderate, resolved types demand more and settle after acceptance. Protected compliance excludes this ordering. In regular feigned restraint, the larger demand is therefore more reassuring: acceptance ends in settlement, while acceptance of the low demand can precede attack.

The same incentive to obtain militarily valuable concessions changes the outcomes of coercion. At maximal pooling demands and a common compliance rate, the target concedes less, which makes settlement less attractive to additional types. In the matched regular

separating comparison, belligerent types leave the high-demand settlement pool and obtain the low-demand concession before attacking. Compliance increases while peaceful transfers decline. War risk and conflict costs rise in both comparisons, reducing total welfare.

Who bears this loss depends on bargaining adjustment. In pooling, the target's smaller concession preserves its resistance payoff and leaves the loss with the coercer. At matched demands and responses, the added option instead benefits belligerent types, and the target loses their gain plus the added conflict costs. Allowing both games to adjust establishes a further limit: under uniform costs, exposure raises minimum attainable war and conflict costs and lowers maximum coercer payoff and total welfare across the characterized pooling and positive two-demand classes. Arbitrary equilibrium pairs can still reverse these rankings.

The mechanism also has precise boundaries. Regular sorting and its matched outcome signs survive any continuous full-support cost distribution. The demand signal becomes less informative about binary resistance resolve, while information about the full cost has no ordering. Removing the concession's military benefit makes maximal-pooling material outcomes coincide, although counting compliance before an unchanged war can still lower peaceful concessions per compliance. The Munich record documents concealed aims and transferred defenses but leaves the strategic low-demand choice unconfirmed. Testing feigned restraint therefore requires evidence about why a demand was chosen as well as what followed its acceptance.

REFERENCES

- Blackwell, David. 1951. "Comparison of Experiments." *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*: 93–102. doi:[10.1525/9780520411586-009](https://doi.org/10.1525/9780520411586-009).
- Cebul, Matthew D., Allan Dafoe, and Nuno P. Monteiro. 2021. "Coercion and the Credibility of Assurances." *The Journal of Politics* 83(3): 975–91. doi:[10.1086/711132](https://doi.org/10.1086/711132).

- Cho, In-Koo, and David M. Kreps. 1987. "Signaling Games and Stable Equilibria." *The Quarterly Journal of Economics* 102(2): 179–221. doi:[10.2307/1885060](https://doi.org/10.2307/1885060).
- Fang, Songying, Jesse C. Johnson, and Brett Ashley Leeds. 2014. "To Concede or to Resist? The Restraining Effect of Military Alliances." *International Organization* 68(4): 775–809. doi:[10.1017/s0020818314000137](https://doi.org/10.1017/s0020818314000137).
- Kydd, Andrew H., and Roseanne W. McManus. 2017. "Threats and Assurances in Crisis Bargaining." *Journal of Conflict Resolution* 61(2): 325–48. doi:[10.1177/0022002715576571](https://doi.org/10.1177/0022002715576571).
- Pauly, Reid B. C. 2024. "Damned If They Do, Damned If They Don't: The Assurance Dilemma in International Coercion." *International Security* 49(1): 91–132. doi:[10.1162/isec_a_00488](https://doi.org/10.1162/isec_a_00488).
- Schelling, Thomas C. 1966. *Arms and Influence*. New Haven, CT: Yale University Press. doi:[10.12987/9780300253481](https://doi.org/10.12987/9780300253481).
- Schram, Peter. 2022. "When Capabilities Backfire: How Improved Hassling Capabilities Produce Worse Outcomes." *The Journal of Politics* 84(4): 2246–60. doi:[10.1086/719637](https://doi.org/10.1086/719637).
- Slantchev, Branislav L. 2010. "Feigning Weakness." *International Organization* 64(3): 357–88. doi:[10.1017/s002081831000010x](https://doi.org/10.1017/s002081831000010x).
- Spaniel, William, and Işıl İdrisoğlu. 2024. "Endogenous Military Strategy and Crisis Bargaining." *Conflict Management and Peace Science* 41(4): 365–91. doi:[10.1177/07388942231180499](https://doi.org/10.1177/07388942231180499).

SUPPORTING INFORMATION

CONTENTS

A. Pooling Equilibrium Formulas	3
B. Preliminary Results	5
C. The Intuitive Criterion and Off-Path Beliefs	6
D. Protected-Compliance Pooling Construction	8
E. Imitation under Positive Compliance	9
F. Exposed-Compliance Pooling Construction	9
G. Failed-Coercion Equilibria	10
H. Pooling Outcome Comparisons	11
I. Independent Compliance Rates at Maximal Pooling Demands	13
J. Two-Demand Sorting Equilibria	13
K. Monotone Two-Pool Sorting	15
L. Proof of the Monotone Characterization	16
L.1. Proof of Proposition 6	16
M. Low Demands Can Feign Restraint Only under Exposed Compliance	17
N. Proof of the Feigned-Restraint Characterization	20
N.1. Proof of Proposition 7	20
O. An Interior Feigned-Restraint Example	21
P. Zero-Demand Sorting Boundaries	22
Q. Proof of Equilibrium-Class Existence	23

Q.1. Proof of Proposition 1	23
R. Two-Pool Outcome Formulas	25
S. Proofs of the Main Outcome Propositions	26
S.1. Proof of Propositions 2–4	26
T. Pure Compliance as a Special Case	28
U. Bargaining Adjustment and Attainable Outcomes	29
U.1. Proof of Proposition 5	30
U.2. Pairwise Reversals with Certain Low-Demand Acceptance	33
V. Distribution, Information, and Zero-Exposure Benchmarks	34
V.1. Regular Sorting under a General Prior	34
V.2. What Demand Signals Reveal	36
V.3. Retaining the Attack Option at Zero Exposure	37
W. Documentary Evidence and Measurement at Munich	40

A. POOLING EQUILIBRIUM FORMULAS

This section collects the algebra behind the pooling results in the main text. A pooling equilibrium has one on-path demand receiving positive compliance; other on-path demands, if any, are surely resisted. Responses to unused demands do not enter this classification. The maximal demands below are maximal within this class. They need not bound the demands in separating equilibria. After resistance, a type's continuation value is

$$R(c) = \max\{p - c, -a\}$$

Under protected compliance, the target accepts a demand $x < p$ when

$$x \leq \frac{h(F(r) - F(t_P(x)))}{1 - F(t_P(x))} \quad (7)$$

where $t_P(x) = p - x$. The maximal accepted demand x_P is the positive root of

$$x^2 + (\bar{c} - 2p - c_2)x - ha = 0 \quad (8)$$

Writing $t_P = p - x_P$, the protected outcomes under pure compliance with the accepted demand are

$$\Omega_P = x_P, \quad T_P = x_P(1 - F(t_P)), \quad q_P = F(t_P) \quad (9)$$

Throughout this section, a pooling outcome written without a compliance-probability argument denotes its pure-compliance endpoint. Conditional coercive ability Ω_j is independent of that probability. For later use, define the expected conflict cost contributed by types below $z \in [0, \bar{c}]$ as

$$I(z) \equiv \int_0^z (c + c_2) dF(c) = \frac{z^2 + 2c_2 z}{2\bar{c}}$$

Thus $K_P = I(t_P)$. At x_P , any $\lambda_P \in (0, 1]$ is supportable, giving

$$\begin{aligned} T_P(\lambda_P) &= \lambda_P x_P (1 - F(t_P)) \\ q_P(\lambda_P) &= F(t_P) + (1 - \lambda_P)(F(r) - F(t_P)) \\ K_P(\lambda_P) &= I(t_P) + (1 - \lambda_P)(I(r) - I(t_P)) \end{aligned}$$

Under exposed compliance, the target's indifference condition for an all-type pool is

$$x(1 - F(t_E(x))) + (h + \gamma x)F(t_E(x)) = hF(r) \quad (10)$$

and the maximal accepted demand x_E is the positive root of

$$(1 - \gamma)^2 x^2 + [\bar{c} - (1 - \gamma)(2p + c_2)]x - ha = 0 \quad (11)$$

Let

$$D_E = 2(1 - \gamma)^2 x_E + \bar{c} - (1 - \gamma)(2p + c_2)$$

$$D_P = 2x_P + \bar{c} - 2p - c_2$$

At the maximal demand, coercive ability and the expected peaceful transfer are

$$\Omega_E = x_E(1 - F(t_E)), \quad T_E(\lambda) = \lambda x_E(1 - F(t_E)) \quad (12)$$

War risk and expected conflict costs are

$$q_E(\lambda) = F(t_E) + (1 - \lambda)(F(r) - F(t_E)) \quad (13)$$

$$K_E(\lambda) = I(t_E) + (1 - \lambda)(I(r) - I(t_E)) \quad (14)$$

Implicit differentiation of the exposed demand equation gives

$$\frac{\partial x_E}{\partial \gamma} = -\frac{x_E[2p + c_2 - 2(1 - \gamma)x_E]}{D_E} < 0 \quad (15)$$

$$\frac{\partial t_E}{\partial \gamma} = x_E - (1 - \gamma)\frac{\partial x_E}{\partial \gamma} > x_E \quad (16)$$

The power derivatives of the two maximal demands are

$$\frac{\partial x_P}{\partial p} = \frac{2x_P + a}{D_P}, \quad \frac{\partial x_E}{\partial p} = \frac{2(1 - \gamma)x_E + a}{D_E} \quad (17)$$

Direct integration gives the coercer's expected payoffs at the maximal pooling demands:

$$U_P = x_P + \frac{t_P^2}{2\bar{c}} \quad (18)$$

$$U_P(\lambda_P) = \lambda_P \left[x_P + \frac{t_P^2}{2\bar{c}} \right] + (1 - \lambda_P) \left[-a + \frac{r^2}{2\bar{c}} \right]$$

$$U_E(\lambda) = \lambda \left[x_E + \frac{t_E^2}{2\bar{c}} \right] + (1 - \lambda) \left[-a + \frac{r^2}{2\bar{c}} \right] \quad (19)$$

For either $j \in \{P, E\}$, the target's expected payoff and total welfare are

$$U_{2,j}(\lambda_j) = 1 - hF(r), \quad V_j(\lambda_j) = 1 - K_j(\lambda_j) - (1 - \lambda_j)a[1 - F(r)]$$

In particular, $V_P = 1 - I(t_P)$ and $V_E = 1 - I(t_E)$ at pure compliance. The main pooling comparison sets $\lambda_P = \lambda_E = \lambda > 0$.

B. PRELIMINARY RESULTS

LEMMA 1 (Continuation cutoffs): With the stated peaceful tie convention, the coercer fights after resistance exactly when $c < r = p + a$. Under exposed compliance, after compliance with demand x , it attacks exactly when $c < t_E(x) = p - (1 - \gamma)x$.

Proof. War after resistance yields $p - c$, while backing down yields $-a$. Attack after compliance yields $p + \gamma x - c$, while settlement yields x . Comparing the relevant payoffs gives both cutoffs. $\square$

LEMMA 2 (Interior accepted demands): Under Assumptions 1–2, x_P and x_E are the unique positive roots of (8) and (11). They satisfy $0 < x_E < x_P < p$ and $0 < t_P < t_E < r < \bar{c}$.

Proof. Under uniformity, the target's benchmark acceptance surplus, multiplied by $\bar{c}$, is

$$h(a + x) - x(\bar{c} - p + x) = ha - (\bar{c} - 2p - c_2)x - x^2$$

Assumption 1 makes this expression positive at zero, negative at p , and strictly decreasing on $[0, p]$. It therefore has the unique root x_P .

For $0 \leq x \leq p$, exposed compliance gives resistance loss minus compliance loss, multiplied by $\bar{c}$, of

$$h(p + a) - (h + \gamma x)[p - (1 - \gamma)x] - x[\bar{c} - p + (1 - \gamma)x]$$

Setting this expression to zero gives (11). Its constant term is negative and its leading coefficient is positive, so it has exactly one positive root. Evaluating the target's acceptance surplus at $x = p$ gives a negative value, which places x_E in $(0, p)$.

Evaluate the exposed-compliance polynomial at x_P . Using (8), its value is

$$\gamma x_P[2p + c_2 - (2 - \gamma)x_P] > 0$$

The last inequality follows from $x_P < p$. Since the polynomial is negative at zero, $x_E < x_P$. Thus

$$t_E - t_P = \gamma x_E + x_P - x_E > 0$$

The remaining inequalities follow from Assumptions 1–2. □

C. THE INTUITIVE CRITERION AND OFF-PATH BELIEFS

Let $v_j(c)$ denote type c 's equilibrium payoff in game $j \in \{P, E\}$. Its payoff from demand d when the target complies with probability λ is

$$u_P(d, c, \lambda) = (1 - \lambda)R(c) + \lambda d,$$

$$u_E(d, c, \lambda) = (1 - \lambda)R(c) + \lambda \max\{d, p + \gamma d - c\}.$$

Continuation play follows Lemma 1. Let $\Lambda_j(d)$ be the set of compliance probabilities that are optimal for the target under some posterior on $[0, \bar{c}]$. After an unused demand d , the types not ruled out by equilibrium dominance are

$$\mathcal{T}_j(d) = \left\{ c \in [0, \bar{c}] : \max_{\lambda \in \Lambda_j(d)} u_j(d, c, \lambda) \geq v_j(c) \right\}.$$

The Intuitive Criterion rejects an equilibrium outcome if some type in $\mathcal{T}_j(d)$ strictly gains from d under every target best response to beliefs supported on $\mathcal{T}_j(d)$ (Cho and Kreps 1987). If this set is empty, the criterion imposes no restriction at d . The following construction supplies one admissible belief and response that deters every type at each unused demand.

LEMMA 3 (Completion under the Intuitive Criterion): Under Assumptions 1–2, an on-path profile with pure demand strategies extends to a PBE satisfying the Intuitive Criterion if its posteriors obey Bayes' rule, its target responses are optimal, no type prefers another used demand, and $v_j(c) \geq R(c)$ for every type. The completion leaves all on-path demands, responses, and payoffs unchanged.

Proof. Fix a game and suppress its subscript. Because $r < \bar{c}$, weak types exist. Choose a demand s used by a weak type and let λ_s be its compliance probability. All weak types face identical payoffs from each demand, so sender optimality gives them the common equilibrium payoff

$$w = \lambda_s s - (1 - \lambda_s)a \leq \lambda_s s.$$

Every type can imitate this demand. Its resistance payoff is at least $-a$, and its compliance payoff is at least s in either game. Consequently, $v(c) \geq w$. Under exposure, a resolved type can also attack after imitating s , so

$$v(c) \geq R(c) + \lambda_s \gamma s \quad \text{for } c < r.$$

These bounds, together with $v(c) \geq R(c)$, suffice to specify all unused demands.

First consider $0 \leq d \leq h$ with $d \geq w$. Compliance is a target best response to belief on a resolved type that would settle after compliance. Such a type exists in both games: under exposure, $\max\{0, t_E(d)\} < r$. Thus acceptance is among the responses used to test equilibrium dominance. It gives every weak type $d \geq w$, so weak types belong to $\mathcal{T}(d)$. Put posterior probability one on a type $c > r$ and resist. The target obtains 1 from resistance and $1 - d$ from compliance, making resistance strictly optimal for $d > 0$ and weakly optimal at zero. Every sender then receives $R(c) \leq v(c)$.

Next consider $0 \leq d < w$. Settlement after compliance gives $d < w \leq v(c)$. If exposure instead induces attack, then $c < t_E(d) \leq p < r$, and the compliance payoff is

$$p + \gamma d - c = R(c) + \gamma d < R(c) + \lambda_s \gamma s \leq v(c).$$

Resistance also gives at most $v(c)$. Hence no pure or mixed target response makes the deviation profitable. If $\mathcal{T}(d)$ is nonempty, put belief on a surviving type and choose a target best response.

If it is empty, use a weak-type belief and resist. Either choice is sequentially rational and deters every sender. This step can require compliance with an unused demand, but that demand remains unchosen because no type gains from it.

Finally, let $d > h$. Against a resolved settler, compliance costs $d > h$ rather than the war loss h . Against a weak type, it costs $d > 0$ rather than zero. Against an exposed attacker, it adds $\gamma d > 0$ to the war loss. Resistance is therefore strictly optimal under every posterior, so $\Lambda_j(d) = \{0\}$. The surviving types, if any, satisfy $v(c) = R(c)$. Put belief on any such type, or use any posterior if no type survives, and resist. Again, no sender gains.

The cases exhaust $[0, 1]$, including zero demands, payoff ties, and arbitrary on-path compliance probabilities. The prescribed beliefs meet the criterion whenever it restricts beliefs, and target responses deter all deviations. Bayes' rule and optimality on path are unchanged, while [Lemma 1](#) supplies optimal continuation play after either target action. This completes the equilibrium. $\square$

D. PROTECTED-COMPLIANCE POOLING CONSTRUCTION

Proof. We construct the full maximal-demand family. Let types $c < t_P$ demand 1 and types $c \geq t_P$ demand x_P . At demand 1, Bayes' rule places probability one on types that fight after resistance. Because $h < 1$ under Assumption 1, the target resists. At x_P , the posterior is the upper tail $c \geq t_P$. Condition (7) binds, so any compliance probability $\lambda_P \in (0, 1]$ is optimal.

Types below t_P receive $R(c) = p - c > x_P$ and lose $\lambda_P[R(c) - x_P]$ by mimicking. Types above t_P gain $\lambda_P[x_P - R(c)] > 0$ from their accepted demand relative to resistance; the cutoff type is indifferent. Thus on-path incentives hold and every type obtains at least $R(c)$. Apply [Lemma 3](#) with $s = x_P$ and $w = \lambda_P x_P - (1 - \lambda_P)a$ to complete the equilibrium.

The supporting beliefs can be explicit here. At an unused $0 \leq d < w$, put belief on $c = 0$ and comply. This type remains eligible because its equilibrium payoff is $R(0)$, and compliance is optimal since $d < w \leq x_P < h$. Every sender receives less than its equilibrium payoff from this accepted deviation. At an unused $w \leq d \leq h$, put belief on a weak type and resist. At $d > h$, put belief on $c = 0$ and resist. If $w \leq 0$, the first case is empty and weak-type beliefs support refusal at zero. These choices satisfy the Intuitive Criterion for every $\lambda_P \in (0, 1]$.

To prove maximality within the one-accepted-demand class, suppose its accepted demand were at least p . Every type would weakly prefer it to resistance, strictly so except possibly at the zero-cost endpoint, so its posterior would be the prior. Acceptance would require $x \leq hF(r) = \frac{hr}{c} < p$, a contradiction to Assumption 1. For an accepted $x < p$, the types with $R(c) < x$ strictly prefer its pool to any resisted demand, while types with $R(c) > x$ prefer resistance. Its posterior is therefore the upper tail $c \geq p - x$, up to the indifferent cutoff. Condition (7) implies $x \leq x_P$. $\square$

E. IMITATION UNDER POSITIVE COMPLIANCE

Proof. Fix a positive demand $x > 0$ that receives compliance with probability $\lambda > 0$ and a type that fights after resistance and attacks after compliance. A surely resisted signal yields $p - c$. Mimicking yields

$$\lambda(p + \gamma x - c) + (1 - \lambda)(p - c) = p - c + \lambda\gamma x > p - c$$

Under Assumption 2, $\lambda\gamma x > 0$, so the type cannot be assigned to the surely resisted signal. The argument uses no restriction on the posterior or on whether the target mixes. A zero demand does not yield the strict gain. $\square$

F. EXPOSED-COMPLIANCE POOLING CONSTRUCTION

Proof. Every type strictly prefers a positive demand receiving positive compliance to a surely resisted signal: attackers gain γx , resolved settlers have $x > p - c$, and weak types have $x > -a$ upon compliance. Thus a pooling equilibrium with one positive accepted demand must place every type there. Compliance costs the target at least x , since an attacking type imposes $h + \gamma x = x + t_E(x) + c_2 > x$, while a settling type imposes x . An accepted all-type pool therefore requires $x \leq hF(r) < p$ under Assumption 1.

Let every type demand $x \in (0, x_E]$. The target's resistance loss minus compliance loss is strictly positive for $x < x_E$ and zero at $x = x_E$. It therefore complies purely below x_E and may comply with any $\lambda \in (0, 1]$ at x_E . The exposed polynomial is strictly increasing on $[0, p]$, so compliance is strictly suboptimal for $x \in (x_E, p)$. The preceding bound excludes $x \geq p$. Thus no positive-compliance pooling equilibrium exists above x_E .

For any demand in this set, a type $c < t_E(x)$ attacks after compliance. Its on-path payoff is $p - c + \lambda\gamma x$, strictly above the $p - c$ from a refused positive deviation. A type $c \in [t_E(x), r)$ settles after compliance and fights after resistance. Because $c \geq p - (1 - \gamma)x$ and $\gamma x > 0$, it has $x > p - c$ and strictly benefits from every positive compliance probability. A type $c \geq r$ obtains $\lambda x + (1 - \lambda)(-a) > -a$. Thus all types prefer the pool to a refused deviation. Below x_E , set $\lambda = 1$; at x_E , use any $\lambda \in (0, 1]$.

All types obtain more than $R(c)$, and on-path sender and target optimality have been verified. Apply Lemma 3 with $s = x$ and $w = \lambda x - (1 - \lambda)a$. In this positive-demand family, weak-type beliefs and resistance work after every unused demand, including zero. For $0 \leq d < w$ and $d > h$, every type is equilibrium dominated, so the criterion imposes no further restriction there. For $w \leq d \leq h$, weak types survive the test and make resistance optimal. Thus the full family satisfies the Intuitive Criterion.

The boundary $x = 0$ also supports an all-type pool. The target strictly prefers compliance because its surplus is $h[F(r) - F(p)] > 0$. Types $c \leq p$ obtain $R(c)$ and are indifferent between the pool and a refused deviation; higher-cost types settle and strictly prefer the pool. Apply Lemma 3 with $s = 0$, $\lambda_s = 1$, and $w = 0$. Weak-type beliefs support resistance at unused $d \leq h$, while belief on $c = 0$, whose payoff equals $R(0)$, supports resistance at $d > h$. This equilibrium satisfies the criterion and has $T = \Omega = 0$, $q = F(p)$, and $K = I(p)$.

Lemma 2 establishes $x_E < x_P$ and (4) at the maximal-demand boundary. Implicitly differentiating (11) with respect to γ gives (15). Its numerator is $x_E[t_E + t_E + c_2] > 0$, and D_E is positive at the positive root, so the derivative is negative. Equation (16) follows immediately. $\square$

G. FAILED-COERCION EQUILIBRIA

Both games admit equilibria in which every demand is resisted. Map types injectively into demands in $(h, 1]$ and let the target resist after inferring the type. If compliance would lead to attack, resistance avoids the additional loss γx . If compliance would lead to settlement, a demand above h costs more than ordinary war. Types above r back down after resistance. Every type receives its resistance continuation value, so none gains by mimicking another resisted signal. Apply Lemma 3 with $\lambda_s = 0$ and $w = -a$. After every unused demand, including

zero, assign belief to a weak type and resist. At $d \leq h$, weak types could strictly gain from acceptance and survive the criterion. At $d > h$, every type survives because refusal is the only rational response and gives its equilibrium payoff. Thus complete coercive failure also satisfies the Intuitive Criterion in both games.

H. POOLING OUTCOME COMPARISONS

Proof. Equations (9) and (12) give

$$\Omega_E = x_E(1 - F(t_E)) < x_E < x_P = \Omega_P$$

At a common $\lambda > 0$, $x_E < x_P$ and $t_E > t_P$ give

$$T_E(\lambda) < \lambda x_P(1 - F(t_P)) = T_P(\lambda)$$

Subtracting the corresponding formulas in the two games gives

$$q_E(\lambda) - q_P(\lambda) = \lambda[F(t_E) - F(t_P)] > 0$$

$$K_E(\lambda) - K_P(\lambda) = \lambda[I(t_E) - I(t_P)] > 0$$

It remains to compare the coercer's expected payoff. At the protected boundary, the target resists against types below t_P and is indifferent between compliance and resistance against the upper tail. At the exposed boundary, it is indifferent between compliance and resistance under the prior. Its ex ante expected payoff is therefore the same in both regimes: the payoff from resisting against the prior.

In either game, total surplus at the maximal pooling demand is

$$V_j(\lambda) = 1 - K_j(\lambda) - (1 - \lambda)a[1 - F(r)], \quad j \in \{P, E\}$$

The last term is the audience cost paid when resistance is followed by backing down. It is identical across games at a common compliance rate. Since the target also receives the same expected payoff,

$$V_E(\lambda) - V_P(\lambda) = U_E(\lambda) - U_P(\lambda) = -\lambda[I(t_E) - I(t_P)] < 0$$

Direct integration gives (18), (19), and the protected mixed-response payoff above. Thus the welfare and payoff comparisons hold for every common $\lambda > 0$.

Implicit differentiation gives (17). Write $s = 1 - \gamma$, $y = sx_E$, and $H = 2p + c_2$. The exposed-demand equation becomes

$$y^2 + \left[\frac{\bar{c}}{s} - H \right] y - ha = 0$$

Thus y is the protected-demand root evaluated at the larger effective support $\frac{\bar{c}}{s}$, so $0 < y < x_P$. Direct simplification gives

$$\frac{\partial x_P}{\partial p} - \frac{\partial x_E}{\partial p} = \frac{2(\bar{c} - H - a)(x_P - y) + \gamma(2x_P + a)(H - 2y)}{D_P D_E}$$

Below $\bar{p}$, both terms in the numerator are positive because $y < x_P < p$. Hence $\frac{\partial x_E}{\partial p} < \frac{\partial x_P}{\partial p}$.

The power derivative of y is $\frac{2y+a}{[2y+\frac{\bar{c}}{s}-H]}$. A second direct comparison gives

$$\frac{\partial x_P}{\partial p} - \frac{\partial y}{\partial p} = \frac{2\left(\frac{\bar{c}}{s} - H - a\right)(x_P - y) + \left(\bar{c}\frac{\gamma}{s}\right)(a + 2y)}{D_P[2y + \frac{\bar{c}}{s} - H]} > 0$$

Since $t_E = p - y$ and $t_P = p - x_P$, this establishes

$$\frac{\partial F(t_E)}{\partial p} > \frac{\partial F(t_P)}{\partial p} > 0$$

below $\bar{p}$.

Therefore

$$\frac{\partial \Omega_E}{\partial p} = \left(\frac{\partial x_E}{\partial p} \right) (1 - F(t_E)) - x_E \left(\frac{\partial F(t_E)}{\partial p} \right) < \frac{\partial x_P}{\partial p} = \frac{\partial \Omega_P}{\partial p}$$

The benchmark derivative is positive. The exposed derivative is also positive. Substituting the exposed-demand equation into its numerator gives

$$\bar{c}(1 - \gamma) \frac{\partial \Omega_E}{\partial p} = \frac{a(\bar{c} - H) + 2y(\bar{c} - p + a) + 3y^2}{2y + \frac{\bar{c}}{1-\gamma} - H} > 0$$

As λ ranges over $(0, 1]$, (13) spans the interval with lower bound $F(t_E)$ and upper limit $F(r)$. Below $\bar{p}$, both bounds rise more rapidly than $F(t_P)$: the lower-bound result follows from the cutoff-derivative comparison above, while $\frac{\partial F(r)}{\partial p} = \frac{1}{\bar{c}} > \frac{\partial F(t_P)}{\partial p}$ gives the upper-bound result.

Likewise, (14) spans the interval with lower bound $I(t_E)$ and upper limit $I(r)$. Below $\bar{p}$, the lower bound rises more rapidly than $I(t_P)$ because both its level and its cutoff derivative are larger. The upper bound does so because $r > t_P$ and $\frac{\partial t_E}{\partial p} < 1$. Finally,

$$\frac{\partial t_E}{\partial p} = \frac{\bar{c} - (1 - \gamma)(2p + c_2 + a)}{D_E}$$

so the exposed lower bounds turn at $\bar{p}_E = \frac{[\frac{\bar{c}}{1-\gamma} - c_2 - a]}{2}$. These are derivatives of the endpoints of the equilibrium ranges. A realized equilibrium with λ_j varying with p also has the response term $-(F(r) - F(t_j)) \frac{d\lambda_j}{dp}$ in its war-risk derivative, and the analogous conflict-cost term. The Intuitive Criterion leaves those response changes undetermined because the entire pooling family satisfies Lemma 3. This proves the pooling parts of Propositions 2–4 without imposing a direction on equilibrium selection as power changes. $\square$

I. INDEPENDENT COMPLIANCE RATES AT MAXIMAL POOLING DEMANDS

Matching compliance probabilities is consequential because the Intuitive Criterion permits their independent selection. For example, let $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, and $\gamma = 0.10$, which satisfy Assumptions 1–2. The maximal demands are $x_P \approx 0.19146$ and $x_E \approx 0.17782$. Choosing $\lambda_P = 0.10$ and $\lambda_E = 1$ gives

$$q_P(\lambda_P) \approx 0.42543 > 0.21998 \approx q_E(\lambda_E), \quad T_P(\lambda_P) \approx 0.01524 < 0.13870 \approx T_E(\lambda_E)$$

$$V_P(\lambda_P) \approx 0.65657 < 0.94721 \approx V_E(\lambda_E)$$

Both equilibria satisfy the Intuitive Criterion by Lemma 3. The common-response comparisons therefore identify assurance effects conditional on the matched response, not an ordering of all equilibria satisfying the criterion.

J. TWO-DEMAND SORTING EQUILIBRIA

This section gives the complete characterization with pure demand strategies and two strictly positive demands, each chosen by a positive mass of types and receiving positive compliance. Let $0 < x < y \leq 1$, and let $\alpha > 0$ and $\beta > 0$ denote the target's compliance probabilities after

y and x . Necessarily, $\alpha < \beta$. To see why, first note that any type assigned to x must obtain a nonnegative compliance gain: it can otherwise demand more than h , induce resistance regardless of the target's belief, and obtain its resistance continuation payoff. Moreover, the compliance gain is strictly increasing in the demand. If $\alpha \geq \beta$, therefore, every type assigned to x would strictly prefer to mimic y . A two-demand equilibrium thus requires $0 < \alpha < \beta \leq 1$. Write $\rho = \frac{\alpha}{\beta} < 1$. For game $j \in \{P, E\}$, define the sender's gain from compliance rather than resistance as

$$G_P(d, c) = \begin{cases} d - p + c & \text{if } c < r \\ d + a & \text{if } c \geq r \end{cases}$$

$$G_E(d, c) = \begin{cases} \max\{\gamma d, d - p + c\} & \text{if } c < r \\ d + a & \text{if } c \geq r. \end{cases}$$

A type chooses y rather than x exactly when

$$\rho G_j(y, c) \geq G_j(x, c) \tag{20}$$

Under protected compliance, compare the gains directly:

$$\rho G_P(y, c) - G_P(x, c) = \begin{cases} \rho y - x + (1 - \rho)(p - c) & \text{if } c < r \\ \rho(y + a) - (x + a) & \text{if } c \geq r. \end{cases}$$

This difference is strictly decreasing below r and constant above it. Its nonnegative set therefore has at most one cutoff. The argument also covers $c_2 = 0$ without dividing by a zero compliance gain. Under exposed compliance, both gains are strictly positive. Define $t_d = p - (1 - \gamma)d$. Then

$$\frac{G_E(x, c)}{G_E(y, c)} = \begin{cases} \frac{x}{y} & \text{if } c \leq t_y \\ \frac{\gamma x}{y - p + c} & \text{if } t_y < c < t_x \\ \frac{x - p + c}{y - p + c} & \text{if } t_x \leq c < r \\ \frac{x + a}{y + a} & \text{if } c \geq r. \end{cases}$$

The ratio is first flat, then decreasing, then increasing, and finally flat, on the branches that intersect the type support. This nonmonotonicity permits either one cutoff or two.

Receiver optimality rules out the upper indifference plateau when $x > 0$. If $\rho \geq \frac{x+a}{y+a}$, every resolved type under protection strictly prefers y . Under exposure, every resolved type that would settle after complying with x strictly prefers y . Thus a low-demand pool could contain only weak types, and under exposure also attackers. Compliance costs the target $x > 0$ against

a weak type and an additional $\gamma x > 0$ against an attacker. The target would strictly resist. Consequently, every equilibrium in this class has $\rho < \frac{x+a}{y+a}$, and its upper cutoff lies below r .

K. MONOTONE TWO-POOL SORTING

In a monotone profile with $0 < x < y \leq 1$, types $c < k$ demand y and types $c \geq k$ demand x . The target complies with probabilities $\alpha < \beta$.

PROPOSITION 6 (Complete monotone characterization): A protected-compliance monotone two-pool equilibrium exists exactly under conditions (i)–(iii):

1. $y = h$ and $0 < k < r$;
2. $\rho = \frac{x-p+k}{h-p+k} = \frac{x-p+k}{k+c_2}$ lies in $(0, 1)$; and
3. $x(\bar{c} - k) \leq h(r - k)$.

An exposed-compliance monotone two-pool equilibrium exists exactly under conditions (iv)–(viii). Let $m_y = \max\{0, t_y\}$. Then

1. $m_y < k < r$ and $t_x < k$;
2. $(h - y)(k - m_y) = \gamma y m_y$;
3. $\rho = \frac{x-p+k}{y-p+k}$ lies in $(0, 1)$;
4. $\rho G_E(y, 0) \geq G_E(x, 0)$; and
5. $x(\bar{c} - k) \leq h(r - k)$.

In either game, a strict final inequality implies $\beta = 1$ and $\alpha = \rho$. At equality, any $\beta \in (0, 1]$ and $\alpha = \rho\beta$ is admissible.

The exposed characterization includes the behaviorally assured equilibrium as a boundary case. If no type attacks after receiving the high demand, then $m_y = 0$, condition (v) forces $y = h$, and the additional exposed incentive restrictions reduce to

$$\gamma \leq \frac{c_2}{h}, \quad \rho c_2 \geq \gamma x \tag{21}$$

These are exactly the protected profiles that survive exposure without changing their on-path behavior. When $m_y > 0$, by contrast, low-cost types demand $y < h$ and attack after compliance.

The target can still mix because settlements by the remaining high-demand types offset those attacks.

L. PROOF OF THE MONOTONE CHARACTERIZATION

L.1. Proof of Proposition 6

Proof. We prove necessity using sender indifference and target optimality, then apply Lemma 3 for sufficiency. Every equilibrium satisfying the criterion is a PBE, so the usual incentive and posterior restrictions remain necessary. Under protected compliance, the gain difference in (20) is strictly decreasing below r . The upper-plateau exclusion above gives $k < r$, and both pools have positive mass, so $k > 0$. Cutoff indifference gives

$$\rho = \frac{x - p + k}{y - p + k}$$

Every type in the high pool fights after resistance and settles after compliance. Since $0 < \alpha < \beta \leq 1$, the target mixes after y , which requires $y = h$. Its cutoff gain $h - p + k = k + c_2$ is strictly positive, so the displayed ratio is well defined. At the low signal, the target's compliance surplus, multiplied by the posterior denominator, is $h(r - k) - x(\bar{c} - k)$, which yields the final inequality. The decreasing gain difference assigns exactly the types below k to y . Types above r strictly prefer x because cutoff indifference implies $(x + a) > \rho(h + a)$. Every on-path gain is nonnegative: $h - p + c = c_2 + c \geq 0$ in the high pool, and $x - p + k > 0$ at the bottom of the low pool. This verifies necessity and sender optimality for sufficiency.

Under exposed compliance, the high-demand pool contains attackers $[0, m_y)$ and settlers $[m_y, k)$. The target's unnormalized payoff difference between compliance and resistance after y is

$$(h - y)(k - m_y) - \gamma y m_y$$

Because $0 < \alpha < 1$, this expression must equal zero. A high pool consisting only of attackers would make it negative, so $m_y < k$. A nondegenerate upper cutoff must lie on the increasing branch of the ratio, giving $t_x < k$ and condition (vi). The ratio is weakly decreasing before t_x and increasing afterward up to r . Thus the bottom type supplies the only additional incentive

constraint, condition (vii), while the cutoff supplies the upper boundary. The low signal contains only settlers after compliance, so target optimality again gives $x(\bar{c} - k) \leq h(r - k)$.

Conversely, conditions (iv)–(viii) make the target indifferent after y and weakly willing to comply after x . The shape of the ratio and the bottom and cutoff inequalities assign every $c < k$ to y and every $c \geq k$ to x . Types $c < m_y$ attack after high-demand compliance, those in $[m_y, k)$ settle, and every type assigned to x settles because $t_x < k$. Exposed compliance gains are strictly positive, and the protected gains were shown to be nonnegative. Both profiles therefore meet Lemma 3, with $s = x$, $\lambda_s = \beta$, and $w = \beta x - (1 - \beta)a$. Its off-path prescription completes both games for every admissible β , including receiver-indifference boundaries. Under protection with $c_2 = 0$, the zero-cost type has $v_P(0) = R(0)$; belief on that type and compliance with an unused $d < w$ satisfy the criterion without creating a profitable deviation. This proves necessity and sufficiency under the stated solution concept. $\square$

M. LOW DEMANDS CAN FEIGN RESTRAINT ONLY UNDER EXPOSED COMPLIANCE

The substantive feature of a two-sided profile with strictly positive demands is that the belligerent types feign restraint with the low demand: an interior interval $[b, k)$ demands y , while the belligerent and weak tails both demand x . This pattern is impossible under protected compliance because the high-demand gain difference decreases with the war cost. It becomes possible only under exposed compliance, which makes acceptance valuable to the most belligerent types as well as to the weakest types.

PROPOSITION 7 (Complete feigned-restraint characterization): With two strictly positive demands, protected compliance has no two-sided two-pool PBE. Exposed compliance has the following two classes.

First, a regular two-sided equilibrium exists if and only if $y = h$, $0 < x < p$, and

$$\rho_{\min} < \rho < \min\{\rho_0, \rho_r\}$$

where

$$t_h = \gamma h - c_2, \quad t_x = p - (1 - \gamma)x$$

$$\rho_{\min} = \frac{\gamma x}{t_x + c_2}, \quad \rho_0 = \frac{\gamma x}{\max\{\gamma h, c_2\}}, \quad \rho_r = \frac{x + a}{h + a}$$

The cutoffs are

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

they satisfy $\max\{0, t_h\} < b < t_x < k < r$, and target optimality requires

$$x(\bar{c} - k + \gamma b) \leq h(r - k) \tag{22}$$

Second, a tie-supported two-sided equilibrium exists if and only if $0 < x < y < h$, $t_y >$

0 , $\rho = \frac{x}{y}$, $k = p$, and

$$b = t_y - \frac{(h - y)(p - t_y)}{\gamma y}$$

lies in $(0, t_y)$, while

$$x(\bar{c} - p) + \gamma x b \leq h a$$

In a canonical representation, types in $[0, b)$ and $[p, \bar{c}]$ demand x , while types in $[b, p)$ demand y . More generally, any measurable subset of $[0, t_y]$ with total length b , hence probability $\frac{b}{\bar{c}}$, can form the lower low-demand pool. Boundary equalities give the corresponding weak and degenerate profiles.

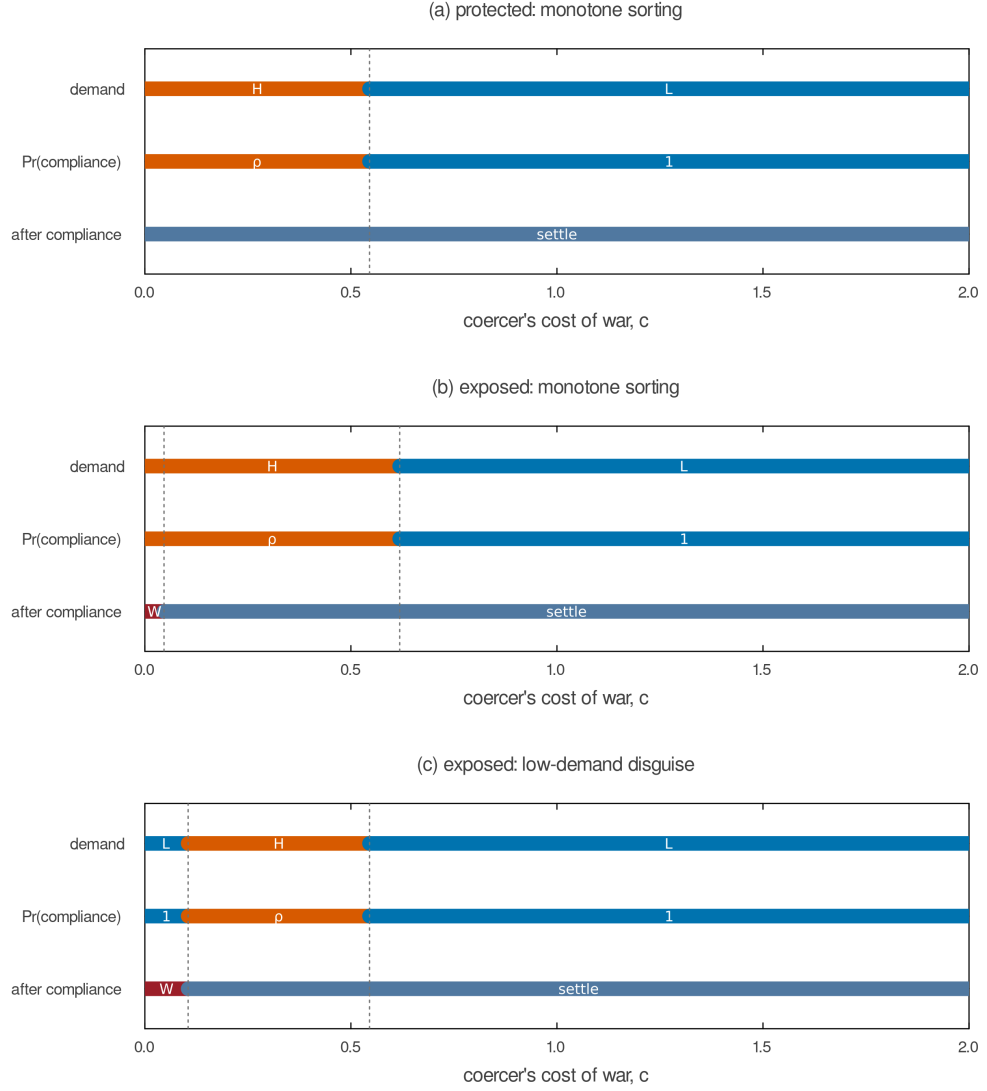


Figure 6: Only exposed compliance lets belligerent types feign restraint with a low demand

Note: Panel (a) shows the monotone ordering required under protected compliance. Panel (b) shows monotone sorting under exposed compliance when the lowest-cost members of the high-demand pool attack after compliance. Panel (c) shows the additional ordering available only under exposed compliance: belligerent and weak types pool at the low demand, leaving moderate types at the high demand. Panels (a) and (c) use $y = h = 0.62$ and $\rho = 0.08$; panel (b) uses $y = 0.615$ and $\rho = 0.187$. Other parameters are $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.10$, $\bar{c} = 2.00$, and $x = 0.10$.

In the regular class, low-cost types below b attack after low-demand compliance, types in $[b, k)$ settle after high-demand compliance, and types above k settle after the low demand. If the target's final inequality is strict, $\beta = 1$ and $\alpha = \rho$; at equality, any $\beta \in (0, 1]$ with $\alpha = \rho\beta$ works.

In the tie-supported class, all types below t_y are indifferent between the two signals. The specified assignment divides this plateau so that the target is willing to mix after y .

N. PROOF OF THE FEIGNED-RESTRAINT CHARACTERIZATION

N.1. Proof of Proposition 7

Proof. Under protected compliance, the gain difference $\rho G_P(y, c) - G_P(x, c)$ is strictly decreasing for $c < r$ and constant above r . The high-demand set is its nonnegative set, which cannot be an interior interval. Two-sided sorting is impossible.

Under exposed compliance, the ratio displayed before Proposition 6 is flat, decreasing, increasing, and flat over its four branches. For a regular two-sided profile, ρ must cross it once on the decreasing branch and once on the increasing branch. The endpoint restrictions are exactly

$$\rho_{\min} < \rho < \min\{\rho_0, \rho_r\}$$

Solving the two equalities yields b and k and implies $\max\{0, t_h\} < b < t_x < k < r$. Because every high-pool type fights after resistance and settles after compliance, target mixing forces $y = h$. At the low signal, attackers $[0, b)$ impose the extra exposure loss $\gamma x b$, settlers $[k, r)$ avert war, and types above r accept a transfer they could otherwise avoid. The target's unnormalized compliance surplus is therefore

$$h(r - k) - x(\bar{c} - k + \gamma b)$$

which proves the regular condition.

The upper flat branch cannot support indifference between two positive-compliance demands, as established before Proposition 6. A positive-mass indifference set can therefore arise only on the bottom flat branch, where the ratio equals $\frac{x}{y}$. Hence $\rho = \frac{x}{y}$ and $t_y > 0$. Above t_x , cutoff indifference gives $k = p$. Let types $[0, b)$ use x and $[b, t_y]$ use y . Target indifference after y requires

$$(h - y)(p - t_y) = \gamma y(t_y - b)$$

which yields the stated b . The target's low-signal compliance condition is $x(\bar{c} - p) + \gamma xb \leq ha$. These restrictions are also sufficient because every type below t_y is indifferent, types in (t_y, p) strictly prefer y , and types above p prefer x .

In both exposed classes, on-path compliance gains are strictly positive. The on-path incentive and target-optimality conditions therefore meet [Lemma 3](#), with $s = x$, $\lambda_s = \beta$, and $w = \beta x - (1 - \beta)a$. Its off-path prescription completes both classes under the Intuitive Criterion, including mixed responses and arbitrary assignments of indifferent bottom types. The protected impossibility follows already from PBE incentives. The strict classes, their boundary equalities, and the positive-mass bottom plateau exhaust the ways a flat-decreasing-increasing-flat ratio can generate a two-sided assignment. $\square$

O. AN INTERIOR FEIGNED-RESTRAINT EXAMPLE

The regular class includes equilibria with demands and acceptance rates bounded away from zero. Set $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, $\bar{c} = 1.50$, $x = 0.20$, and $\rho = 0.20$, with $y = h = 0.62$. All maintained assumptions hold, and the regular conditions are strict:

$$\rho_{\min} = \frac{1}{8} < \rho = \frac{1}{5} < \rho_0 = \frac{10}{31} < \rho_r = \frac{25}{46}$$

$$0 < t_h = 0.166 < b = 0.28 < t_x = 0.46 < k = 0.505 < r = 0.90$$

The target's unnormalized surplus from accepting the low demand is

$$h(r - k) - x(\bar{c} - k + \gamma b) = 0.0291 > 0$$

so $\beta = 1$ and $\alpha = \rho$. The belligerent low-demand group, high-demand group, and upper low-demand group have masses $\frac{14}{75}$, $\frac{3}{20}$, and $\frac{199}{300}$, respectively. The upper low-demand group includes weak types with $c \geq r$, whose population mass is $\frac{2}{5}$.

For these parameters, the difference between the coercer's expected payoffs from the high and low demands is

$$D_P(c) = \begin{cases} 0.404 - 0.8c & \text{if } c < 0.90 \\ -0.316 & \text{if } c \geq 0.90 \end{cases}$$

under protection, and

$$D_E(c) = \begin{cases} -0.0228 & \text{if } c \leq 0.166 \\ -0.056 + 0.2c & \text{if } 0.166 < c < 0.46 \\ 0.404 - 0.8c & \text{if } 0.46 \leq c < 0.90 \\ -0.316 & \text{if } c \geq 0.90 \end{cases}$$

under exposure. The protected profile assigns types below k to the high demand; the exposed profile assigns only types between b and k to it. The target is indifferent after the high demand in both games. Its unnormalized low-demand compliance surplus under protection is $0.0459 > 0$, so the same demands and acceptance rates support the protected counterpart. Every type strictly exceeds its resistance payoff, and [Lemma 3](#) completes both profiles with resistance after unused demands.

The example extends to an open region. Allow independent perturbations of at most 0.001 in each of $p, c_2, a, \gamma, \bar{c}, x, \rho$, while setting $y = p + c_2$, $\alpha = \rho$, and $\beta = 1$. Rational interval bounds preserve all maintained assumptions and regular-class inequalities throughout this box. In particular, $x, \rho \geq 0.199$, the exposed high-demand group has mass greater than 0.14428, and the exposed low-demand compliance surplus exceeds 0.02264. The interior of the box therefore supplies an open set with substantial demands, acceptance rates, and group masses. This is a local existence result, with no claim about their magnitudes throughout the parameter space.

At the center of the box, the matched comparison gives

$$\begin{aligned} \Delta P_C &= \frac{56}{375}, & \Delta T &= -\frac{217}{9375}, & \Delta q &= \frac{14}{375} \\ \Delta U &= \frac{49}{9375}, & \Delta U_2 &= -\frac{7}{625}, & \Delta V &= -\frac{56}{9375} \end{aligned}$$

Thus exposure increases compliance by about 15 percentage points while reducing the peaceful transfer. Belligerent types gain from the added attack option, and the target loses both this gain and the additional joint conflict costs.

P. ZERO-DEMAND SORTING BOUNDARIES

The restriction to strictly positive demands matters because the target can accept a zero demand from weak types. For example, let $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, and $\gamma = 0.10$, with $x = 0$, $y = 0.61$, $\beta = 1$, and $\alpha = \rho = \frac{30}{91} = \frac{a}{y+a}$. In the protected game, set $k_P = \frac{279}{305} \approx$

$0.91475 > r = 0.90$. In the exposed game, set $k_E = \frac{554379}{610000} \approx 0.90882 > r$, and note that $t_y = 0.051$. In each game, assign $c < k_j$ to y and $c \geq k_j$ to zero. All resolved types strictly prefer y ; every weak type is indifferent between the two demands, receiving zero from either. The target is indifferent after the high demand because

$$yk_P = hr, \quad yk_E + (t_y + c_2)t_y = hr$$

It is also indifferent after the zero demand, whose pool contains only weak types. Every type receives at least $R(c)$, so Lemma 3 applies with $s = 0$, $\lambda_s = 1$, and $w = 0$. Weak-type beliefs support resistance at unused $d \leq h$. At $d > h$, every type receives strictly more than $R(c)$ on path, so the criterion is vacuous and weak-type beliefs again support resistance. These equilibria satisfy the Intuitive Criterion and have an upper indifference plateau and $k_j > r$; they lie outside the positive-demand characterizations above.

Q. PROOF OF EQUILIBRIUM-CLASS EXISTENCE

Q.1. Proof of Proposition 1

Proof. The proof proceeds by construction and then establishes necessity for reassuring separation. Lemma 2 and the protected-compliance pooling construction give a positive-compliance pool at x_P for every admissible parameter vector. Lemma 2 and the exposed-compliance pooling construction give an all-type pool at every $x \in (0, x_E]$. Thus pooling exists throughout the maintained parameter space in both games.

For protected monotone sorting, set $y = h$, $k = p$, and $\rho = \frac{x}{h}$. Choose

$$0 < x < \min\left\{h, \frac{ha}{\bar{c} - p}\right\}$$

Then $0 < k < r$, $\rho \in (0, 1)$, and

$$\rho = \frac{x - p + k}{h - p + k} = \frac{x}{h}, \quad x(\bar{c} - k) < h(r - k) = ha$$

These are the protected conditions in Proposition 6, so a reassuring monotone two-pool equilibrium exists. The protected gain difference is decreasing below r and constant above it. Its high-demand set can therefore have only one cutoff, which rules out feigned restraint.

Consider next a reassuring monotone equilibrium under exposed compliance. Because no high-demand type attacks after compliance, $m_y = 0$. Target indifference after the high demand and Proposition 6 imply $(h - y)k = 0$. Since $k > 0$, $y = h$. No attack after high-demand compliance then requires

$$t_h = p - (1 - \gamma)h = \gamma h - c_2 \leq 0$$

or equivalently $\gamma \leq \frac{c_2}{h}$. This proves necessity. For sufficiency, suppose $\gamma \leq \frac{c_2}{h}$ and again set $y = h$, $k = p$, and $\rho = \frac{x}{h}$, with x satisfying the strict bound above. Then $m_y = 0$, $t_x < k$, and

$$\rho G_E(h, 0) = \frac{xc_2}{h} \geq \gamma x = G_E(x, 0)$$

The remaining exposed conditions in Proposition 6 coincide with the protected conditions just verified. Hence reassuring monotone sorting exists if and only if $\gamma \leq \frac{c_2}{h}$.

It remains to construct a feigned-restraint equilibrium for every admissible $\gamma > 0$. Let

$$L = \max\{0, \gamma h - c_2\}$$

Because $\gamma < 1$, $L < p$. Choose $b \in (L, p)$. For $x > 0$, define

$$\rho = \frac{\gamma x}{b + c_2}, \quad t_x = p - (1 - \gamma)x, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

This choice makes $b = \gamma \frac{x}{\rho} - c_2$. Since $b > L$, it also gives

$$b + c_2 > \max\{\gamma h, c_2\}$$

and therefore $\rho < \rho_0$. As x approaches zero, ρ approaches zero, t_x and k approach p , ρ_r approaches $\frac{a}{h+a} > 0$, and

$$h(r - k) - x(\bar{c} - k + \gamma b)$$

approaches $ha > 0$. Since $b < p < r$, continuity permits a sufficiently small $x > 0$ such that $b < t_x$, $\rho < \min\{1, \rho_r\}$, $k < r$, and the low-demand compliance inequality is strict. Moreover,

$$\rho = \frac{\gamma x}{b + c_2} > \frac{\gamma x}{t_x + c_2} = \rho_{\min}$$

and this last inequality is equivalent to $t_x < k$. All strict conditions for the regular feigned-restraint equilibrium in Proposition 7 therefore hold. Such an equilibrium exists at every admissible parameter vector with $\gamma > 0$.

Finally, the demand-choice condition assigns a type to the high demand when $\rho G_j(y, c) \geq G_j(x, c)$. Under protection, the gain difference has one crossing. Under exposure, the gain ratio is flat, decreasing, increasing, and flat; its sublevel set at ρ is either a lower interval or an interior interval. Receiver optimality excludes indifference on the upper plateau, while bottom-plateau ties are covered by Proposition 7. Only indifferent types can be assigned in other ways. The two orderings therefore exhaust the class with pure demand strategies, two strictly positive demands, positive mass at both signals, and positive compliance at each. Every existence construction above satisfies the Intuitive Criterion by Lemma 3, while each impossibility follows from necessary PBE conditions. Thus the equilibrium-class characterization holds under the stated solution concept. $\square$

R. TWO-POOL OUTCOME FORMULAS

Let P_C denote the ex ante probability of compliance. In a monotone equilibrium, let m_y be the length of the interval of high-demand types that attack after compliance; its prior probability is $\frac{m_y}{\bar{c}}$. Then

$$P_C^M = \frac{\alpha k + \beta(\bar{c} - k)}{\bar{c}}$$

$$T_M = \frac{\alpha y(k - m_y) + \beta x(\bar{c} - k)}{\bar{c}}, \quad \Omega_M = \frac{T_M}{P_C^M}$$

$$q_M = \frac{m_y + (1 - \alpha)(k - m_y) + (1 - \beta)(r - k)}{\bar{c}}$$

$$K_M = I(m_y) + (1 - \alpha)(I(k) - I(m_y)) + (1 - \beta)(I(r) - I(k))$$

For protected compliance, set $m_y = 0$ and $y = h$. For the behaviorally assured exposed subclass, set $m_y = 0$ as well.

For a regular two-sided equilibrium,

$$\begin{aligned}
P_C^T &= \frac{\alpha(k-b) + \beta(b+\bar{c}-k)}{\bar{c}} \\
T_T &= \frac{\alpha h(k-b) + \beta x(\bar{c}-k)}{\bar{c}}, \quad \Omega_T = \frac{T_T}{P_C^T} \\
q_T &= \frac{b + (1-\alpha)(k-b) + (1-\beta)(r-k)}{\bar{c}} \\
K_T &= I(b) + (1-\alpha)(I(k) - I(b)) + (1-\beta)(I(r) - I(k))
\end{aligned}$$

For the tie-supported class, replace k by p in P_C^T , use

$$\begin{aligned}
T_T &= \frac{\alpha y(p-t_y) + \beta x(\bar{c}-p)}{\bar{c}} \\
q_T &= \frac{t_y + (1-\alpha)(p-t_y) + (1-\beta)(r-p)}{\bar{c}} \\
K_T &= I(t_y) + (1-\alpha)(I(p) - I(t_y)) + (1-\beta)(I(r) - I(p))
\end{aligned}$$

S. PROOFS OF THE MAIN OUTCOME PROPOSITIONS

S.1. Proof of Propositions 2–4

Proof. The pooling comparison above establishes the first part of each proposition. For separation, Proposition 6 establishes that protected monotone sorting requires $y = h$. Under exposed monotone sorting, target indifference requires $(h-y)(k-m_y) = \gamma y m_y$, with $k > m_y$. Hence $y = h$ when $m_y = 0$ and $y < h$ exactly when $m_y > 0$.

Proposition 7 proves that protected compliance has no two-sided two-pool PBE and gives necessary and sufficient conditions for the exposed regular class. This class contains an open set. For example, at $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, $\gamma = 0.10$, $x = 0.10$, $y = h = 0.62$, and $\rho = 0.08$, all of Proposition 7's inequalities are strict: $0.042 < b = 0.105 < t_x = 0.51 < k = 0.545 < r = 0.90$, and the low-demand compliance inequality has left side 0.147 and right side 0.220. Continuity preserves the inequalities in a neighborhood of these parameters.

In every regular exposed equilibrium, the two sender-indifference conditions imply

$$\rho = \frac{\gamma x}{c_2 + b} \quad \wedge \quad \rho = \frac{x - p + k}{c_2 + k}$$

Solving gives

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

Holding x and ρ fixed while remaining in the regular branch gives $d_{\frac{b}{d}}\gamma = \frac{x}{\rho} > 0$ and $d_{\frac{k}{d}}\gamma = 0$.

Let A denote post-compliance attack. Then

$$\Pr(A \text{ mid } x, C) = \frac{b}{b + \bar{c} - k}, \quad \frac{d \Pr(A \text{ mid } x, C)}{d\gamma} = \text{frac}\{x(\bar{c} - k), \rho(b + \bar{c} - k)^2\} > 0$$

The left side of the target's low-demand compliance condition (22) has derivative

$$\frac{d}{d\gamma} x(\bar{c} - k + \gamma b) = x \left(b + \frac{\gamma x}{\rho} \right) > 0$$

while its right side is constant. Thus greater exposure makes the low demand less reassuring and its compliance condition harder to satisfy.

For the outcome comparison, take a regular exposed two-sided equilibrium in which the low-demand inequality is strict, so $\beta = 1$ and $\alpha = \rho$. Its protected counterpart uses the same h, x, ρ , and upper cutoff k . Under protection, the high-demand settlement set is $\mathcal{H}_P = [0, k)$. Under exposed compliance it is $\mathcal{H}_E = [b, k)$. Because types are uniform on $[0, \bar{c}]$, their probability masses are $\frac{k}{\bar{c}}$ and $\frac{k-b}{\bar{c}}$. The exposed mass therefore falls at rate

$$\frac{d}{d\gamma} \frac{k - b}{\bar{c}} = -\frac{x}{\rho \bar{c}} < 0$$

The types removed from this set enter the low-demand pool and attack after compliance. The target is indifferent after h in both games. At the low demand, removing the exposure loss $\gamma x b$ makes compliance strictly optimal under protection, so the same target responses are sequentially rational.

Let Δ denote the exposed outcome minus the protected outcome. Only types $c \in [0, b)$ change signals. Using the formulas above gives

$$\Delta P_C = \frac{(1 - \rho)b}{\bar{c}} > 0, \quad \Delta T = -\frac{\rho h b}{\bar{c}} < 0$$

$$\Delta q = \frac{\rho b}{\bar{c}} > 0, \quad \Delta K = \rho I(b) > 0$$

For coercive ability, define

$$L_P = \bar{c} - (1 - \rho)k, \quad N_P = \rho h k + x(\bar{c} - k)$$

and $L_E = L_P + (1 - \rho)b$, $N_E = N_P - \rho h b$. Then

$$\Delta \Omega = -\frac{b(\rho h L_P + (1 - \rho)N_P)}{L_P L_E} < 0$$

Under protection, type $c < b$ receives $p - c + \rho(c + c_2)$. Under exposure it receives $p - c + \gamma x$.

The lower cutoff condition gives $\gamma x = \rho(b + c_2)$. Integration therefore yields

$$\Delta U = \int_0^b \rho(b - c) dF(c) = \frac{\rho b^2}{2\bar{c}} > 0$$

The target's payoff changes by

$$\Delta U_2 = -\frac{\gamma x b}{\bar{c}} = -\frac{\rho b(b + c_2)}{\bar{c}} < 0$$

Hence the change in total surplus is

$$\Delta(U + U_2) = -\rho I(b) < 0$$

These calculations prove the separating parts of Propositions 2–4. Both members of each matched comparison satisfy the Intuitive Criterion by [Lemma 3](#), whose completion changes no on-path payoff or response. Hence all reported comparisons hold under the stated solution concept. $\square$

T. PURE COMPLIANCE AS A SPECIAL CASE

Set $\lambda_P = \lambda_E = 1$ at the maximal pooling demands. Then $q_j = F(t_j)$ and $K_j = I(t_j)$ for $j \in \{P, E\}$. The protected and exposed cutoff derivatives are

$$\frac{\partial q_P}{\partial p} = \frac{\bar{c} - 2p - c_2 - a}{\bar{c} D_P}, \quad \frac{\partial q_E}{\partial p} = \frac{\bar{c} - (1 - \gamma)(2p + c_2 + a)}{\bar{c} D_E}$$

They generate distinct turning points:

$$\bar{p}_P = \frac{\bar{c} - c_2 - a}{2}, \quad \bar{p}_E = \frac{\frac{\bar{c}}{1-\gamma} - c_2 - a}{2} > \bar{p}_P \quad (23)$$

Below $\bar{p}_P$, war rises faster under exposed than protected compliance. Between the two thresholds, protected-compliance war falls while exposed-compliance war continues to rise. Above $\bar{p}_E$, both derivatives are negative. Expected conflict cost inherits these signs because $K_j = I(t_j)$ and $I'(t) > 0$; below $\bar{p}_P$, it also rises faster under exposed compliance because $t_E > t_P$ and $\frac{\partial t_E}{\partial p} > \frac{\partial t_P}{\partial p} > 0$.

U. BARGAINING ADJUSTMENT AND ATTAINABLE OUTCOMES

This section proves [Proposition 5](#) and characterizes the marginal outcome ranges when demands and target responses can adjust in both games. The result maintains the uniform prior. It covers every equilibrium characterized in [Proposition 6](#) and [Proposition 7](#), including the exposed monotone and tie-supported classes and their admissible mixed responses. A marginal range describes one outcome at a time; the ranges do not specify which combinations are jointly attainable.

Write $g_P = 0$, $g_E = \gamma$, and $t_j(x) = p - (1 - g_j)x$. Define

$$\Psi_j(x) = hr - \bar{c}x - t_j(x)[t_j(x) + c_2] = ha - [\bar{c} - (1 - g_j)(2p + c_2)]x - (1 - g_j)^2 x^2$$

Its unique positive zero is the maximal pooling demand x_j . The maintained assumptions imply $0 < x_E < x_P < p$ and $0 < t_P < t_E < r < \bar{c}$. The favorable pooling endpoints are

$$q_j^* = F(t_j), \quad K_j^* = I(t_j), \quad U_j^* = hF(r) - I(t_j), \quad V_j^* = 1 - I(t_j)$$

where $t_j = t_j(x_j)$. Universal resistance yields

$$q_R = F(r), \quad K_R = I(r), \quad U_R = -a + \frac{r^2}{2\bar{c}}, \quad V_R = 1 - I(r) - a[1 - F(r)]$$

Within the class of pure demand strategies using two strictly positive demands, each chosen by positive mass and receiving positive compliance, the exact marginal ranges in game j are

$$q \in (q_j^*, q_R), \quad K \in (K_j^*, K_R), \quad U \in (U_R, U_j^*), \quad V \in (V_R, V_j^*)$$

Every value in each interval is attainable, and no endpoint is attained within this class. Adding the characterized pooling and all-resisted equilibria closes the four intervals. Because $t_E > t_P$, each exposed interval is a strict subset of its protected counterpart. This is a statement about attainable outcomes, without a rule selecting an equilibrium.

U.1. Proof of Proposition 5

Proof. First take a two-demand profile with certain low-demand acceptance, $\beta = 1$, and high-demand acceptance $\alpha = \rho \in (0, 1)$. In every characterized class, weak types choose the low demand x and settle, and high-demand types are resolved. Backing down therefore never occurs on path.

Let $v(c)$ be the coercer's equilibrium payoff and $w(c)$ its probability of war. Each demand's continuation payoff is a convex, piecewise affine function of the cost. Sender optimality makes v the maximum of these functions, so v is absolutely continuous and $v'(c) = -w(c)$ almost everywhere. In the bottom indifference plateau, both demands induce war and have the same slope; its arbitrary assignment does not affect this conclusion. Since $v(\bar{c}) = x$, integration under the uniform prior gives

$$\bar{c}q = v(0) - x, \quad U = x + \frac{1}{\bar{c}} \int_0^{\bar{c}} cw(c) dc = x + K - c_2q$$

Under protection, the zero-cost type can guarantee resistance and war by demanding more than h , so $v(0) \geq p$. Under exposure, it can imitate the surely accepted low demand and then attack, so $v(0) \geq p + \gamma x$. Thus, writing $z = \bar{c}q$,

$$z \geq t_j(x)$$

At a fixed war probability, conflict costs are minimized by concentrating war among the lowest-cost types. Specifically,

$$\bar{c}[K - I(z)] = \int_0^{\bar{c}} (c - z)[w(c) - \mathbb{1}[c < z]] dc \geq 0$$

Both factors have the same sign below and above z . Equality requires $w(c) = \mathbb{1}[c < z]$ almost everywhere. Every positive high-demand pool contains a positive mass of resolved settlers: the

target would not mix after a demand whose entire pool attacked after compliance. These settlers fight with probability $1 - \rho \in (0, 1)$, so equality is impossible and $K > I(z)$.

The target can resist either demand, giving

$$U_2 \geq 1 - hF(r)$$

Every coercer can imitate the low demand and obtain at least x . Together with $U + U_2 = 1 - K$, this implies $x \leq h\frac{r}{c} < p$, and hence $t_j(x) > 0$. Combining the envelope identity, the target's option to resist, and the strict conflict-cost bound yields

$$hr \geq \bar{c}(U + K) = \bar{c}x + 2\bar{c}K - c_2z > \bar{c}x + z^2 + c_2z \geq \bar{c}x + t_j(x)^2 + c_2t_j(x)$$

Therefore $\Psi_j(x) > 0$. Assumption 1 makes Ψ_j strictly decreasing on the nonnegative axis, so $x < x_j$. It follows that

$$q \geq \frac{t_j(x)}{\bar{c}} > q_j^*, \quad K > I(\bar{c}q) > K_j^*, \quad U \leq hF(r) - K < U_j^*, \quad V = 1 - K < V_j^*$$

War occurs only among resolved types, and high-demand settlers avoid it with positive probability. Hence $q < q_R$ and $K < K_R$. Every type obtains at least its resistance payoff, and the positive mass of weak types strictly gains, so $U > U_R$. Total welfare also strictly exceeds V_R .

If the original equilibrium has $\beta < 1$, the target is indifferent after both demands. Raising β to one and retaining $\rho = \frac{\alpha}{\beta}$ preserves its best responses and multiplies all between-demand sender gains by the same positive factor. The resulting profile satisfies [Lemma 3](#). For each outcome $Y \in \{q, K, U, V\}$, the original profile satisfies

$$Y(\beta) = \beta Y(1) + (1 - \beta)Y_R$$

Its outcome lies strictly between the pure-low-response outcome and universal resistance. This proves all strict bounds, including mixed low-demand responses.

To establish sharpness and the entire marginal ranges, construct two-demand equilibria with low-demand indifference approaching the favorable pooling endpoint. Under protection, choose $k > t_P$ approaching t_P , and set

$$x = \frac{h(r - k)}{\bar{c} - k}, \quad \rho = \frac{x - p + k}{k + c_2}, \quad y = h$$

At the limit, $x = x_P$ and $\rho = 0$. Multiplying the numerator of ρ by $\bar{c} - k$ gives

$$H(k) = hr - p\bar{c} + (\bar{c} - c_2)k - k^2$$

Here $H(t_P) = 0$ and $H'(t_P) = \bar{c} - c_2 - 2t_P > 0$. Thus for k sufficiently close to t_P , the protected characterization holds with $0 < \rho < 1$. These equilibria approach the maximal pool, while their positive high-demand acceptance excludes the limit from the two-demand class.

Under exposure, choose $x < x_E$ approaching x_E , put $t = t_E(x)$ and $s = \Psi_E(x) > 0$, and set

$$\delta = \frac{-s + \sqrt{s^2 + 4\gamma x s(h - x)}}{2\gamma x}, \quad b = t - \delta, \quad \rho = \frac{\gamma x}{b + c_2}, \quad k = t + \frac{\rho\delta}{1 - \rho}, \quad y = h$$

The root satisfies $\gamma x \delta^2 = s(h - x - \delta)$ and tends to zero. Since $t_E > \max\{0, t_h\}$ and $t_E < r$, all strict regular incentive restrictions hold sufficiently close to the limit. The target's low-demand surplus is

$$h(r - k) - x(\bar{c} - k + \gamma b) = \Psi_E(x) - (t - b)(k - t) = s - \frac{\rho\delta^2}{1 - \rho} = 0$$

The last equality uses $\frac{\rho}{1 - \rho} = \gamma \frac{x}{h - x - \delta}$. Thus these are regular equilibria with low-demand indifference. At $\beta = 1$,

$$\bar{c}q = t, \quad K = I(t) + \frac{\rho\delta^2}{2\bar{c}(1 - \rho)} \rightarrow I(t_E)$$

Target indifference gives $U_2 = 1 - hF(r)$, so U and V also approach their pooling endpoints. The high-demand group has positive mass along the sequence but vanishes in the limit. Every finite member satisfies the Intuitive Criterion by [Lemma 3](#).

At each sequence member, any $\beta \in (0, 1]$ is admissible. Varying β fills the interval between its pure-low-response outcome and universal resistance for each scalar outcome. Taking sequence members sufficiently close to the pooling limit reaches every interior value in each stated interval.

Finally, pure pooling war and conflict costs decrease as the accepted demand increases, whereas $U(x) = x + \frac{t_j(x)^2}{2\bar{c}}$ has derivative $1 - \frac{(1 - g_j)t_j(x)}{\bar{c}} > 0$. Mixed pooling responses combine these outcomes with universal resistance. Thus no characterized pooling equilibrium exceeds the favorable endpoint or the opposite resistance endpoint. Maximal pooling with certain

compliance and all-resisted equilibria attain them, respectively. This closes the four marginal ranges and proves the common extremal comparison. $\square$

U.2. Pairwise Reversals with Certain Low-Demand Acceptance

The nested marginal ranges do not rank arbitrary equilibrium pairs. At every admissible parameter vector, one game's equilibria can approach its favorable endpoint while the other's approach universal resistance. This gives both directions of comparison for q, K, U, V .

Reversals also occur with $\beta = 1$ and strict low-demand acceptance incentives. Set $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, and $\bar{c} = 1.50$, with $y = h = 0.62$ in all four profiles. Consider protected profiles with $(x, k, \rho) = (0.01, 0.60, \frac{1}{62})$ and $(0.30, 0.32, \frac{1}{17})$. Their unnormalized low-demand surpluses are 0.177 and 0.0056. The first exposed profile is the interior example in [Supporting Information O](#). The second has $x = 0.01$, $\rho = \frac{3}{220}$, $b = 0.20$, and $k = \frac{6493}{10850}$, also with strictly positive low-demand surplus. All four satisfy their characterization conditions.

Comparing the interior exposed profile with the first protected profile gives

$$\begin{aligned} \Delta P_C &= \frac{212}{775} > 0, \quad \Delta T = \frac{2119}{15000} > 0, \quad \Delta q = -\frac{202}{2325} < 0 \\ \Delta K &= -\frac{43309}{930000} < 0, \quad \Delta U = \frac{135007}{930000} > 0, \quad \Delta V = \frac{43309}{930000} > 0 \end{aligned}$$

Comparing the second exposed profile with the second protected profile reverses all six signs, and the sign of $\Delta\Omega$ reverses as well. Strict inequalities and continuity preserve these comparisons in neighborhoods of the primitives and the independently chosen menus. Thus the reversals do not require low-demand compliance to approach zero.

The joint outcomes also distinguish the games. Under protection, compliance and war are disjoint, so $P_C + q \leq 1$. In regular exposed sorting with $\beta = 1$,

$$P_C + q = 1 + \frac{b}{\bar{c}} > 1$$

No protected equilibrium can reproduce that pair of compliance and war probabilities. The marginal nesting of q, K, U, V therefore does not imply replication of joint outcomes.

V. DISTRIBUTION, INFORMATION, AND ZERO-EXPOSURE BENCHMARKS

The results in this section replace uniformity by an atomless, strictly increasing CDF F with full support on $[0, \bar{c}]$. The remaining maintained assumptions continue to hold, except that the final benchmark sets $\gamma = 0$. Write $I(z) = \int_0^z (c + c_2) dF(c)$. These results concern regular sorting and maximal pooling at zero exposure; they do not extend the uniform-prior outcome bounds in [Proposition 5](#).

V.1. Regular Sorting under a General Prior

LEMMA 4 (Regular feigned restraint and its matched comparisons survive a general prior): A regular exposed profile has $y = h$, $0 < x < h$, $0 < \rho < 1$, and

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}, \quad \max\{0, t_h\} < b < t_x < k < r$$

Its remaining acceptance constraint is

$$x[1 - F(k) + \gamma F(b)] \leq h[F(r) - F(k)]$$

These conditions are necessary and sufficient, with $\beta = 1$ at strict inequality and any $\beta \in (0, 1]$ at equality. Such profiles exist for every admissible $\gamma > 0$. At $\beta = 1$, the same demands and responses support a protected monotone counterpart. Both profiles satisfy the Intuitive Criterion, and their outcome differences are

$$\Delta P_C = (1 - \rho)F(b) > 0, \quad \Delta T = -\rho h F(b) < 0$$

$$\Delta q = \rho F(b) > 0, \quad \Delta K = \rho I(b) > 0$$

$$\Delta U = \rho \int_0^b (b - c) dF(c) > 0, \quad \Delta U_2 = -\rho(b + c_2)F(b) < 0, \quad \Delta V = -\rho I(b) < 0$$

Peaceful concessions per compliance, Ω , also fall.

Proof. Demand incentives are pointwise in the cost. The ratio $\frac{G_E(x, c)}{G_E(h, c)}$ is constant, decreasing, increasing, and constant on the branches displayed in the feigned-restraint characterization. Its

two strict crossings with ρ determine the same b and k under every prior. The high-demand group consists of resolved settlers. Since the target mixes there, its indifference requires $y = h$. In the low-demand group, attackers below b generate an acceptance loss γx , resolved settlers between k and r generate a gain $h - x$, and weak settlers above r generate a loss x . The unnormalized acceptance surplus is therefore

$$S_F = h[F(r) - F(k)] - x[1 - F(k) + \gamma F(b)]$$

This proves the characterization and the response restrictions.

For existence, choose a fixed $b_0 \in (\max\{0, \gamma h - c_2\}, p)$ and set

$$\rho(x) = \frac{\gamma x}{b_0 + c_2}, \quad k(x) = \frac{p - x + \rho(x)c_2}{1 - \rho(x)}$$

As x tends to zero from above, $\rho(x)$ tends to zero, t_x and $k(x)$ tend to p , and

$$k(x) - t_x = \frac{\rho(x)(t_x - b_0)}{1 - \rho(x)} > 0$$

for all sufficiently small positive x . All strict cutoff restrictions then hold. Continuity and full support give

$$S_F \rightarrow h[F(r) - F(p)] > 0$$

so the low demand is surely accepted. No density or hazard-rate restriction is required.

Under protection, the same h, x, ρ have a single sender crossing at k . The low-demand acceptance surplus becomes $S_F + \gamma x F(b) > 0$, while high-demand indifference is unchanged. This proves the matched counterpart at $\beta = 1$. An exposed profile with $\beta < 1$ cannot be matched at the same responses, because the protected target strictly prefers low-demand compliance.

The off-path construction in [Lemma 3](#) does not use uniformity: sender equilibrium dominance is pointwise, and the on-path posterior is the same F conditioned on its demand cells. In particular, weak types share the low-demand payoff $w = \beta x - (1 - \beta)a$, and all types can imitate it. For an unused $d \in [0, h]$ with $d \geq w$, weak types survive the dominance test; a posterior on surviving weak types and resistance deters deviation. If $d < w$, settlement yields less than the imitable payoff. An attacking type also gains less from d than from the used low

demand because $d < w \leq \beta x$ and $\gamma > 0$. Thus no response creates a profitable deviation there. A posterior on surviving types and a receiver best response completes this case. For $d > h$, resistance is strictly optimal under every posterior. The exceptional zero-cost protected survivor when $c_2 = 0$ is treated as in Lemma 3. Both profiles therefore satisfy the same refinement under the changed prior.

Only types below b change behavior in the matched pair. Under protection, they demand h , settle with probability ρ , and fight otherwise. Under exposure, they obtain x certainly and attack. Their changes in compliance, peaceful transfer, and war probability are $1 - \rho$, $-\rho h$, and ρ . Their payoff change is

$$p + \gamma x - c - [p - c + \rho(c + c_2)] = \rho(b - c)$$

using $\gamma x = \rho(b + c_2)$. The target loses γx for each such type. Integrating proves every displayed identity. Since $T_P > 0$, $\Delta T < 0$, and $\Delta P_C > 0$,

$$\Delta \Omega = \frac{\Delta T - \Omega_P \Delta P_C}{P_{C,E}} < 0$$

□

The prior can change which menus satisfy the target's constraint. For example, at $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, and $\bar{c} = 1.50$, replacing uniformity by $F(c) = \left(\frac{c}{1.50}\right)^2$ makes the interior menu $x = \rho = 0.20$ infeasible: $S_F = -\frac{39743}{1500000} < 0$. The menu $x = \rho = 0.10$ instead satisfies $S_F = \frac{114481}{2278125} > 0$. Existence under every prior therefore allows demands and responses to adjust. The uniform restriction $x < p$ is also unnecessary for general-prior regular sorting; the lemma uses $x < h$ together with the cutoff restrictions.

V.2. What Demand Signals Reveal

LEMMA 5 (Exposure reduces information about binary resistance resolve): Fix F and a matched regular pair from Lemma 4. For $Z = \mathbb{1}[c < r]$, the exposed demand signal is a strict Blackwell garbling of the protected signal. Relabel a protected high demand

as low with probability $\frac{F(b)}{F(k)}$ and leave a protected low demand unchanged. The two demand experiments are Blackwell incomparable about the full cost c .

Proof. Use the comparison of experiments in Blackwell (1951). Let $A = F(k)$, $B = F(b)$, and $\pi = F(r)$, so $0 < B < A < \pi < 1$. Conditional on $Z = 0$, both games always produce the low demand. Conditional on $Z = 1$, their high-demand probabilities are $\frac{A}{\pi}$ and $\frac{A-B}{\pi}$. The stated kernel reduces the first probability to the second without conditioning on the realized binary state. A reverse kernel must keep an exposed low demand low, because it is the only signal in state $Z = 0$. It can therefore produce a high demand in state $Z = 1$ with probability at most $\frac{A-B}{\pi} < \frac{A}{\pi}$. The binary order is strict.

For the full cost, protection partitions the support into $[0, k)$ and $[k, \bar{c}]$, while exposure partitions it into $[b, k)$ and $[0, b) \cup [k, \bar{c}]$. A garbling of the protected high signal cannot send it to low for $c < b$ and to high for $b < c < k$. A reverse garbling of the exposed low signal cannot send it to high for $c < b$ and to low for $c > k$. All three intervals have positive prior mass, so neither garbling exists. Equivalently, protection identifies $\mathbb{1}[c < k]$ perfectly, whereas exposure identifies $\mathbb{1}[b < c < k]$ perfectly; each classification has positive error under the other experiment. $\square$

The binary comparison averages over the fixed prior's costs within each resolve state. Its kernel depends on $\frac{F(b)}{F(k)}$ and is not a common kernel for arbitrary changes in that within-state distribution. Nor does the result rank information about willingness to stop after compliance. That decision depends on the demand and attack technology. The target's actual payoffs also depend on these features, so binary informativeness alone does not sign its equilibrium payoff.

V.3. Retaining the Attack Option at Zero Exposure

LEMMA 6 (Zero exposure preserves maximal-pooling material outcomes): Set $\gamma = 0$ while retaining the exposed game's attack option. Let $t(x) = \max\{p - x, 0\}$ and

$$\Psi_F(x) = h[F(r) - F(t(x))] - x[1 - F(t(x))]$$

Both games have the same maximal pooling demand

$$x^* = \max\{x \in [0, 1] : \Psi_F(x) \geq 0\} \in (0, h), \quad t^* = t(x^*)$$

with $\Psi_F(x^*) = 0$. Every common compliance rate $\lambda \in (0, 1]$ is supportable at this demand. Terminal settlement, war, and backing-down probabilities and payoffs agree type by type almost surely. Consequently $(T, q, K, U, U_2, V)_E = (T, q, K, U, U_2, V)_P$.

In the exposed game, any prior mass $m \in [0, F(t^*)]$ of attacking types may join the accepted pool. Thus

$$P_{C,P} = \lambda[1 - F(t^*)], \quad P_{C,E} = \lambda[1 - F(t^*) + m]$$

$$\Omega_P = x^*, \quad \Omega_E = x^* \frac{1 - F(t^*)}{1 - F(t^*) + m}$$

A strict gap in Ω occurs exactly when $m > 0$. Regular feigned restraint with strict demand preferences disappears at $\gamma = 0$.

Proof. Under protection, types above $t(x)$ strictly prefer an accepted positive demand to resistance, and types below it prefer resistance. Under exposure at $\gamma = 0$, the former still join the accepted pool. The latter earn $p - c$ from either acceptance followed by attack or resistance followed by war. They are indifferent about joining. Full support and atomlessness permit any accepted-attacker mass between zero and $F(t(x))$.

An accepted attacker gives the target identical compliance and resistance payoffs. The remaining accepted types generate total acceptance surplus $\Psi_F(x)$ in either game. The different posterior denominators do not change its sign. Thus $\Psi_F(x) \geq 0$ characterizes demands feasible at some positive compliance rate. Strict inequality requires $\lambda = 1$; equality permits every positive $\lambda \leq 1$. Assign excluded attackers to demand 1, which is surely resisted because $h < 1$. On-path incentives hold as stated. The completion in [Lemma 3](#) extends to zero exposure by retaining attacking types that are indifferent in its survivor set: for $d < w$, attacking at an unused demand yields $R(c) \leq v(c)$, so no deviation is profitable even when dominance is weak. The remaining off-path cases are unchanged.

Continuity gives $\Psi_F(0) = h[F(r) - F(p)] > 0$ and $\Psi_F(h) = h[F(r) - 1] < 0$. No $x \geq h$ is feasible. The largest feasible demand therefore exists in $(0, h)$ and binds the target's constraint. This proves common maximality and independence from λ without requiring a unique positive root.

At x^* , types below t^* fight surely in both games. Above t^* , the common acceptance rate gives the same settlement, war, and backing-down lotteries. Since the concession is not retained separately after war, changing the history from resistance and war to compliance and attack changes no material payoff at $\gamma = 0$. This proves typewise equivalence and the aggregate equality. Counting accepted attackers adds λm to compliance without changing the peaceful transfer $\lambda x^*[1 - F(t^*)]$, which gives the formulas for Ω .

Finally, in regular feigned restraint the high-demand pool consists of resolved settlers, so target mixing requires $y = h$. A positive-cost low-demand attacker gains zero from compliance with x but gains $c + c_2 > 0$ from compliance with h . Positive high-demand compliance therefore rules out its strict preference for the low demand. The cutoff equation $\rho(b + c_2) = \gamma x = 0$ cannot hold with $\rho, b > 0$. $\square$

Under uniformity and Assumption 1, $x^* = x_P < p$ and $t^* > 0$. The all-type exposed pool sets $m = F(t^*)$ and has $\Omega_E = x^*[1 - F(t^*)] < \Omega_P$, despite equal material outcomes. It is one admissible composition; setting $m = 0$ makes the compliance counts coincide too. For a general prior, if $hF(r) \geq p$, the common maximal demand is $x^* = hF(r)$ and $t^* = 0$, so no strict composition effect is possible. If $hF(r) < p$, then $x^* < p$. These alternatives follow from $\Psi_F(x) = hF(r) - x$ for $x \geq p$.

The zero-exposure result does not equate complete equilibrium correspondences. Types that would attack after either demand can be indifferent between multiple positive demands. Such assignments can survive at $\gamma = 0$ even though the strict incentives supporting regular feigned restraint disappear. The statement also excludes $\lambda = 0$, at which coercive ability conditional on compliance is undefined.

W. DOCUMENTARY EVIDENCE AND MEASUREMENT AT MUNICH

The application separates evidence produced before the outcome from later reconstructions. The pre-settlement record consists of a German military directive, Chamberlain's meeting minute and parliamentary statement, Hitler's public speech, diplomatic reports, and the agreement itself. The October records concern actual transfers and subsequent operational planning. The March meeting record and diplomatic reporting establish the form of the later coercion. German documents were created contemporaneously but are read here in postwar English translations; their documentary provenance should not be confused with a contemporaneously published Allied account. US diplomatic reports often relay identified offices or unnamed informants and are not independent observations of every asserted fact. The British archival packet supplies reproductions and transcripts; its educational narrative is not treated as evidence.^[5]

The coding is deliberately narrower than assigning equilibrium types. A claim is coded as limited when the coercer publicly bounds the requested territorial revision; this does not assign a normalized demand share or a military value. Prior expansive aims are identified from the pre-settlement directive, while exposure is identified from the agreement's preservation term and reports of the transferred defenses. Acceptance is dated by implementation, not by assuming Czechoslovakia signed the four-power agreement. No private cost, posterior probability, target mixing rate, or weak-type comparison is observed. A statement by Chamberlain is coded as the broker's public interpretation, not as the target's belief. The relevant post-compliance outcome is further military coercion and occupation; it is not coded as the model's costly battle merely because coercion involved troops.

The strongest rival explanations concern demand choice and acceptance. A transfer that improves a predator's position can arise through coercive maximalism or attempts to provoke rejection, without strategic selection of a low demand. Allied pressure and expectations about future preparedness can induce acceptance despite distrust. The application therefore supports two observable links while withholding the full mechanism claim. Establishing feigned-restraint sorting empirically would require independent evidence that the coercer chose the less resisted

^[5]Chamberlain's minute of the meeting at Berchtesgaden, 15 September 1938, FO 371/21738, reproduced by The National Archives, pp. 8–10.

demand because compliance conferred military advantage, together with comparisons capable of distinguishing belligerent and weak senders at that demand from resolved senders who demand more and stop. That evidence is absent from this application.